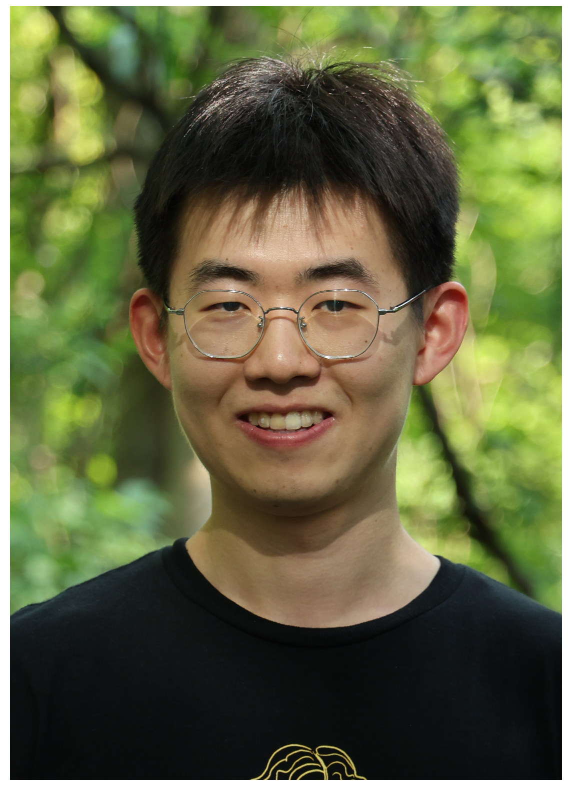
Retrocausal capacity of a quantum channel

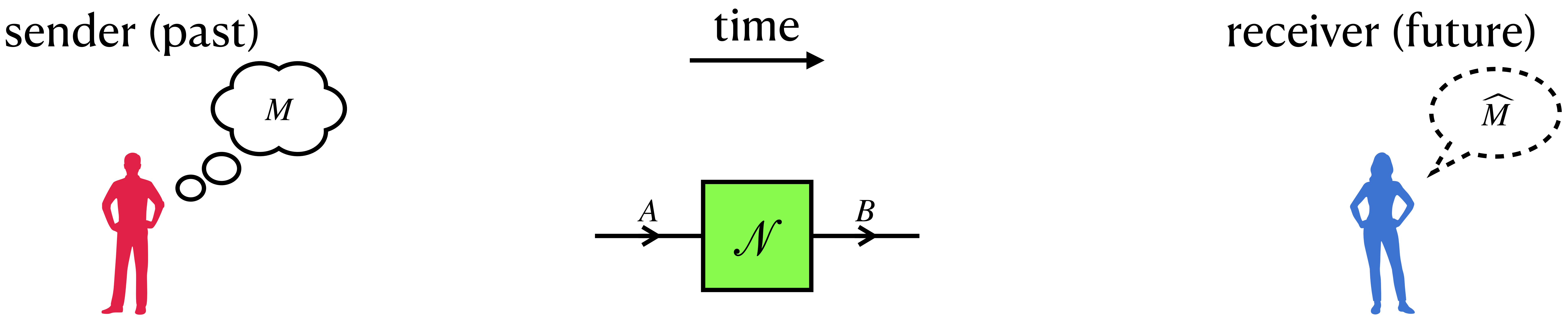
Kaiyuan Ji (Cornell), Seth Lloyd (MIT), Mark M. Wilde (Cornell)



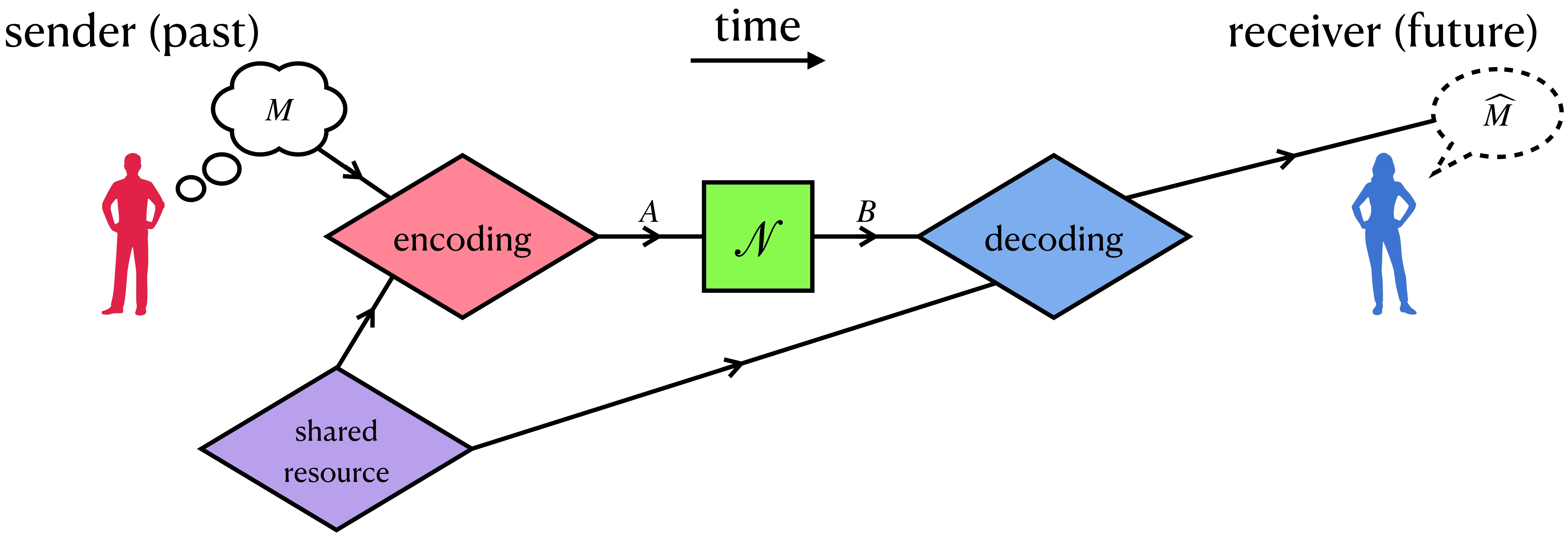
arXiv:2509.08965

Quantum Resources 2026, Tokyo

Communication



Communication



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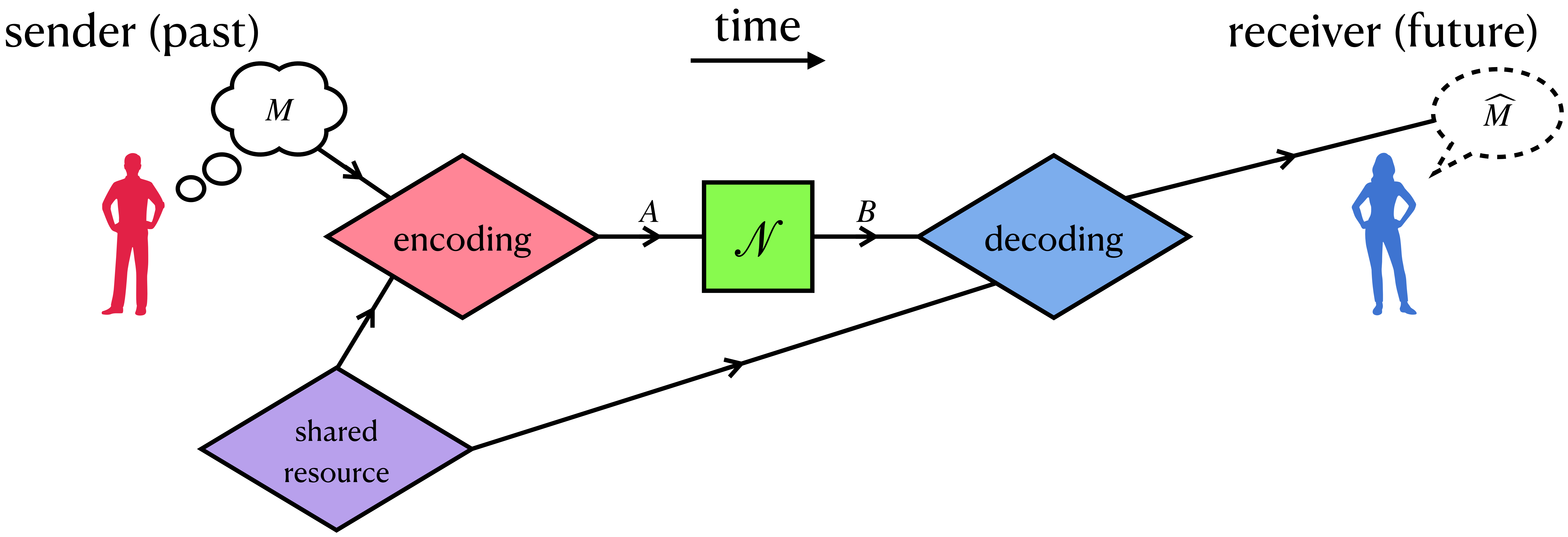
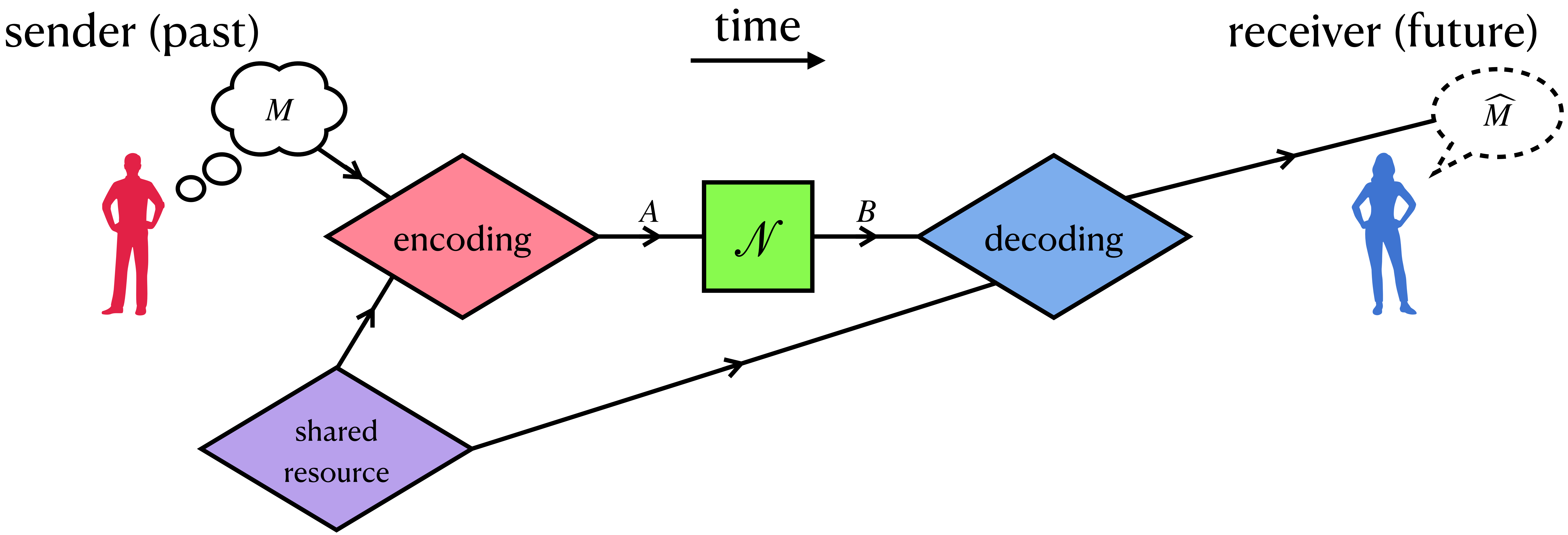


Quantum Shannon theory studies the fundamental limit of communication (forward in time) through a given noisy channel (CPTP map) $\mathcal{N}_{A\to B}$

Communication

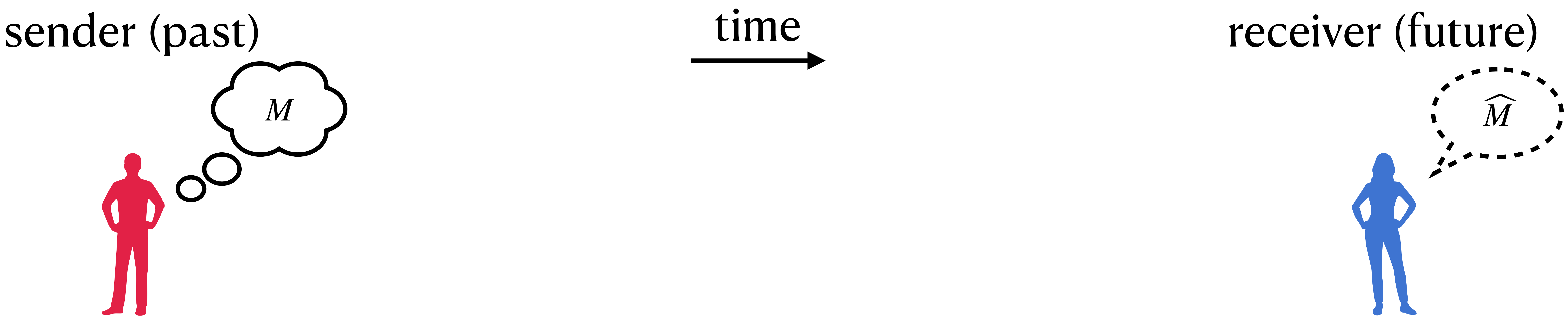


Quantum Shannon theory studies the fundamental limit of communication (forward in time) through a given noisy channel (CPTP map) $\mathcal{N}_{A\to B}$

- "Capacity" of $\mathcal{N}_{A\to B}$

Holevo, Probl. Peredachi Inf. 9, 3 (1973)
Schumacher, PRA 54, 2614 (1996)
...
...

Communication



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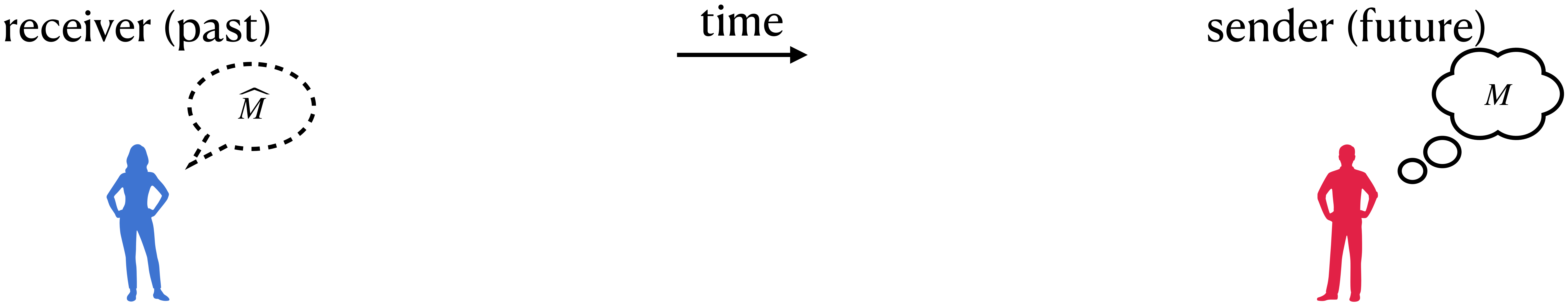
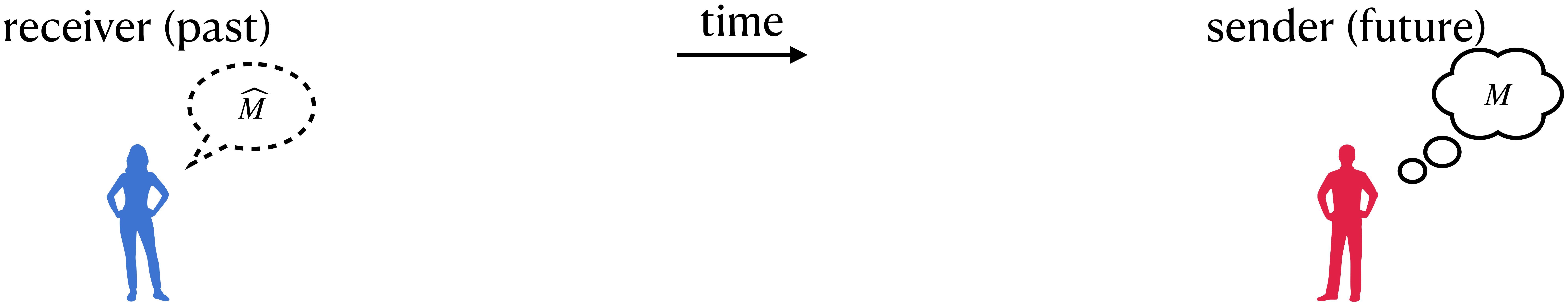


Communication, but backward in time!



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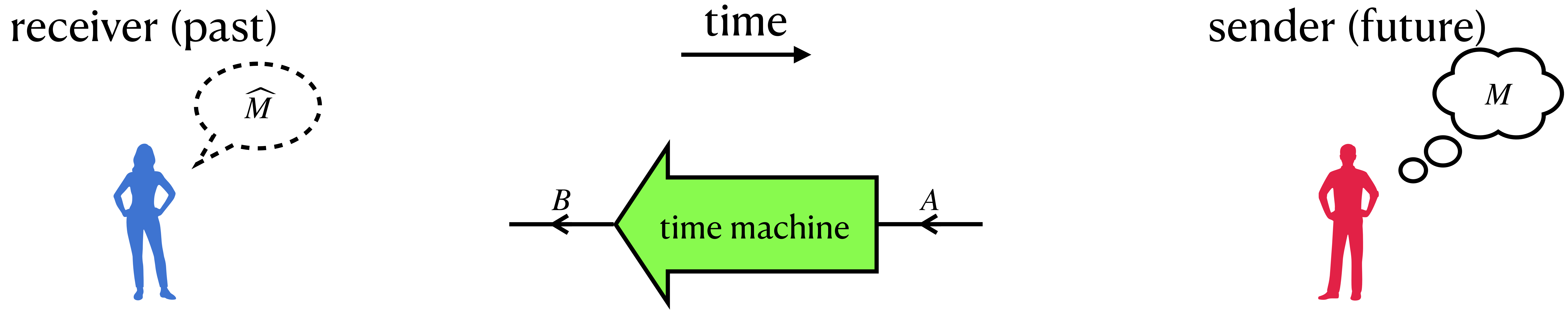

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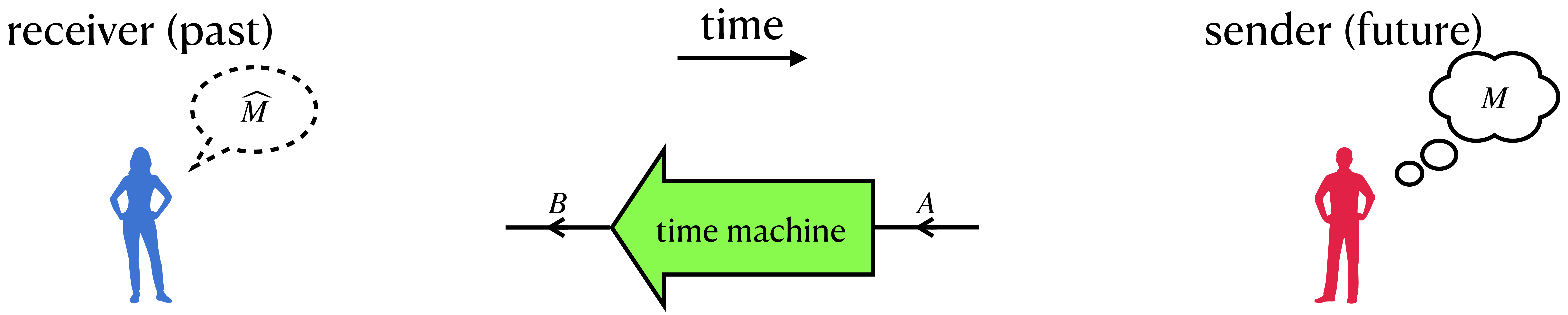

We study the fundamental limit of communication, backward in time, through a given **noisy time machine** (assuming its existence)

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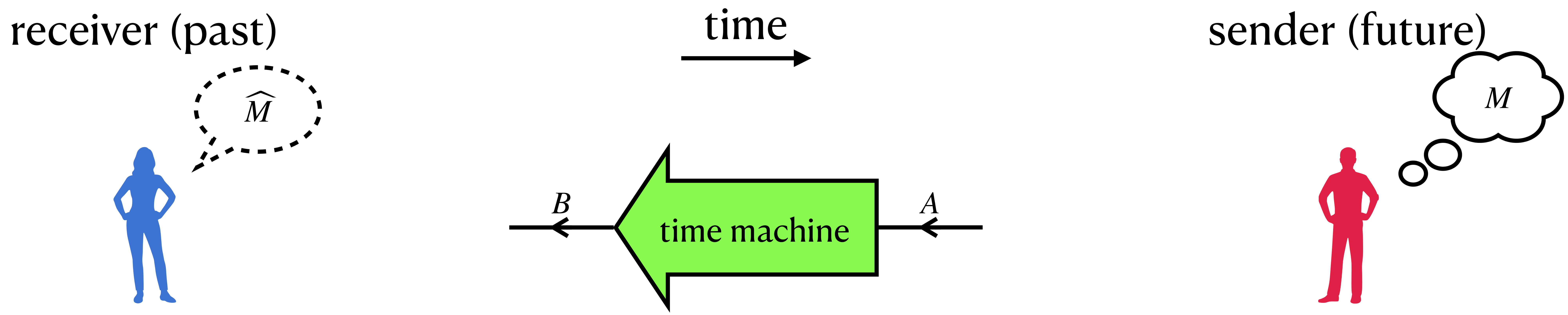

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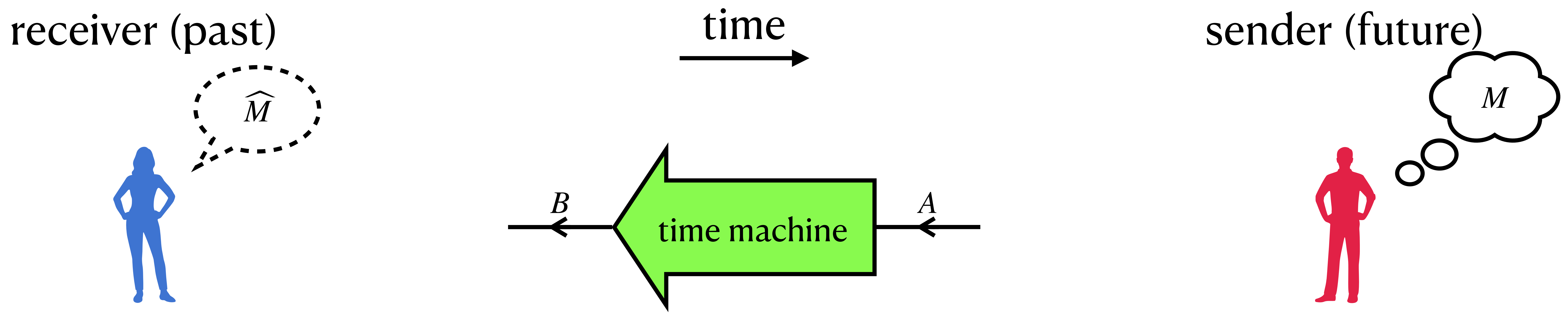

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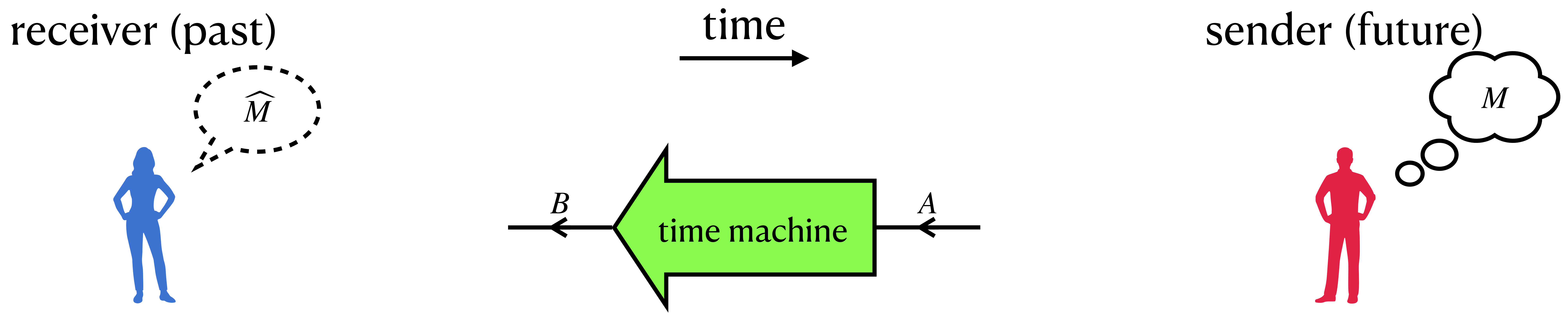

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We solve this precisely in the one-shot regime

How to describe a noisy time machine mathematically?

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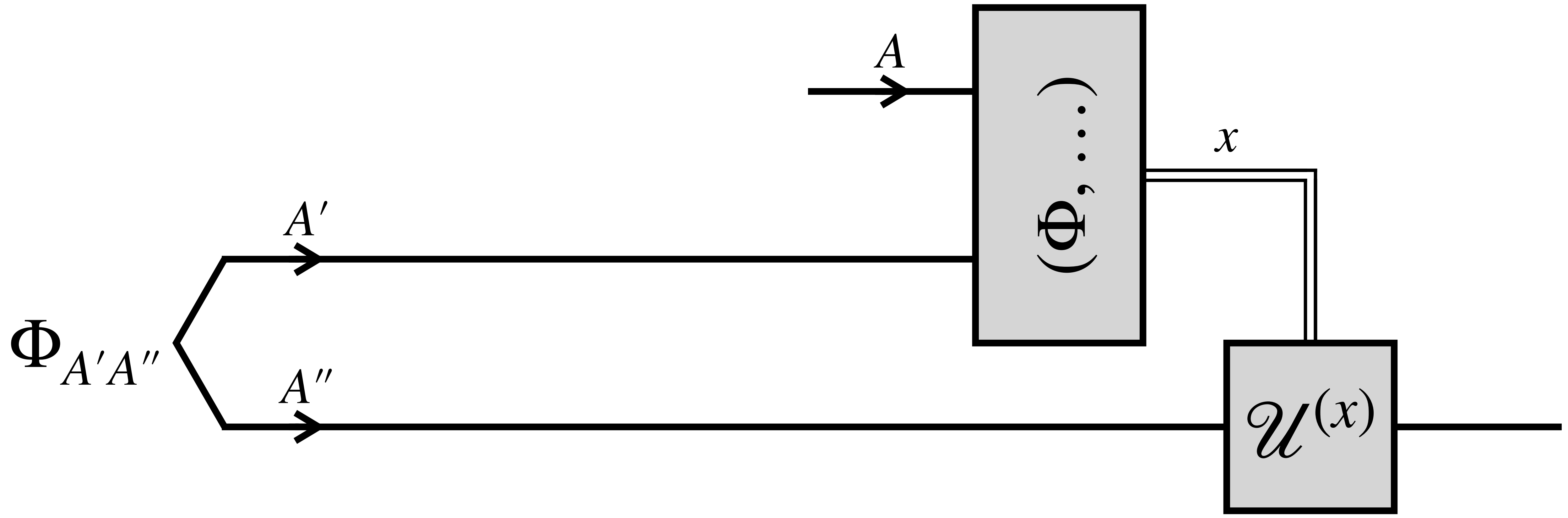
Teleportation



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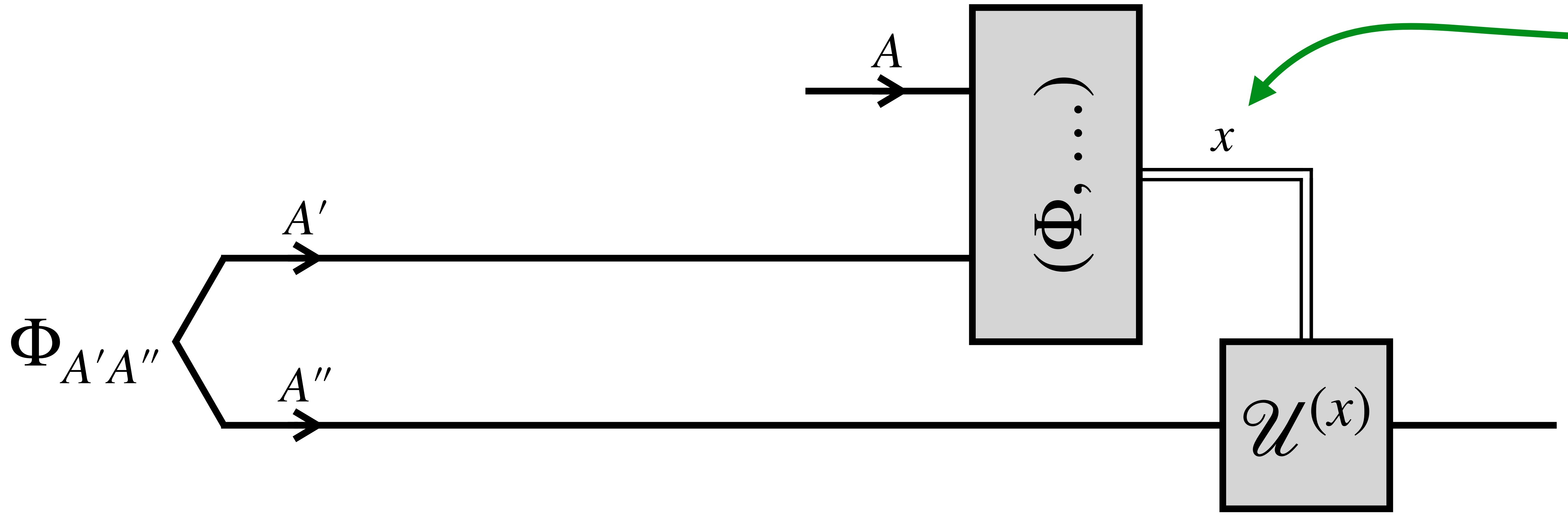
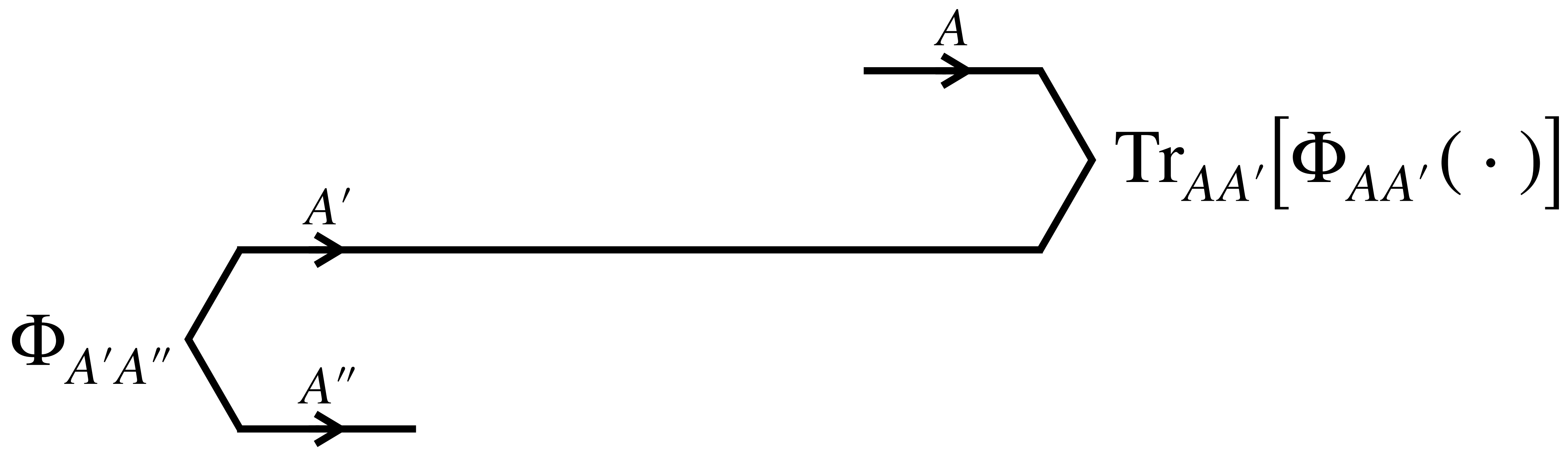


assume $x = 0$ (projection onto $\Phi_{AA'}$)

How to describe a noisy time machine mathematically?



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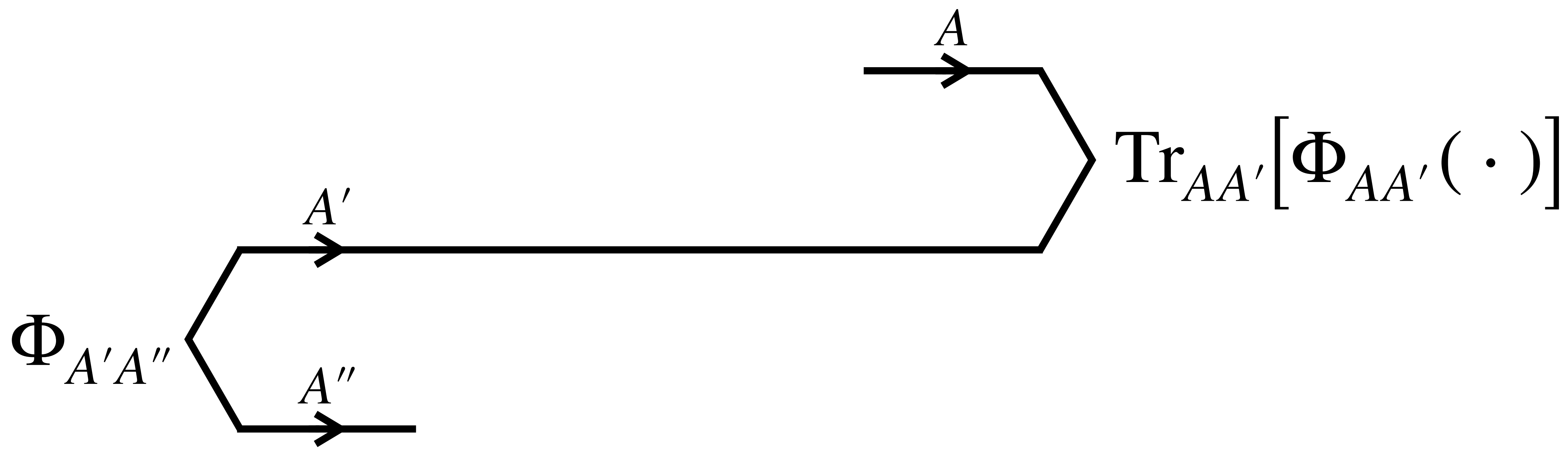
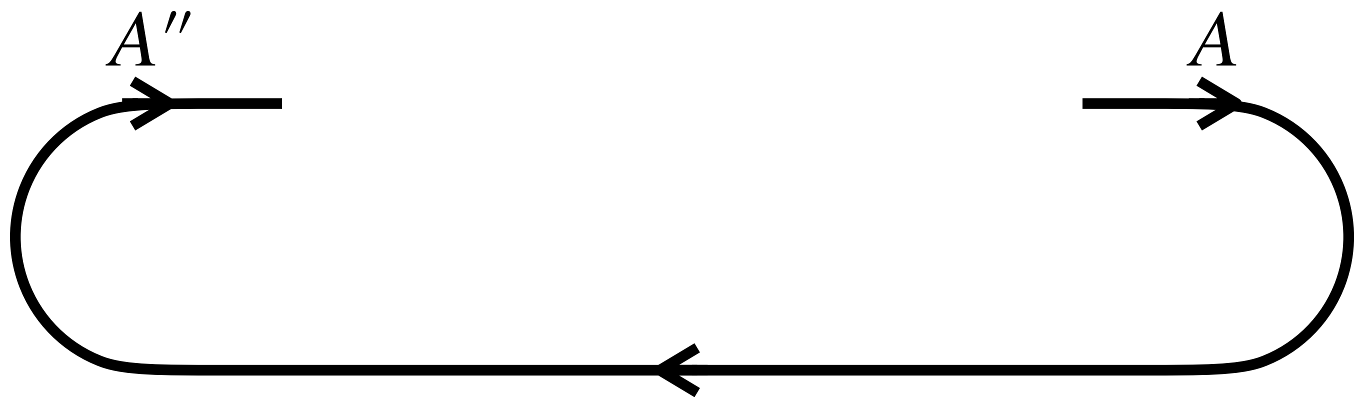
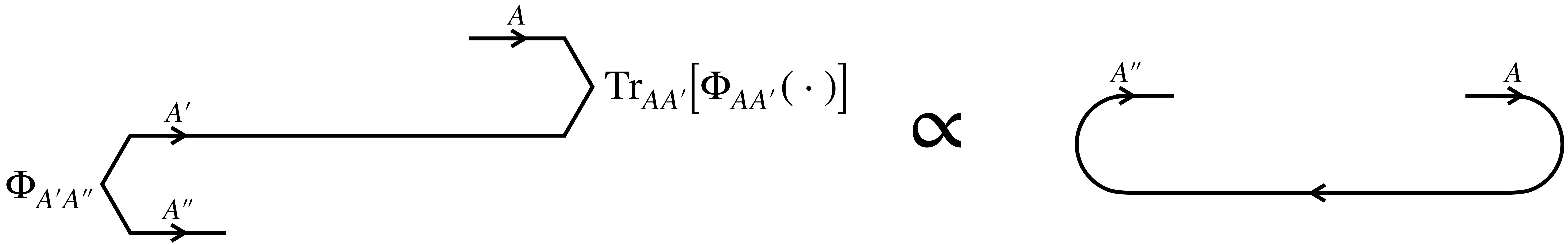


$\propto$



How to describe a noisy time machine mathematically?

Postselected closed timelike curve (P-CTC)



Bennett and Schumacher, unpublished (2002)
Svetlichny, arXiv:0902.4898 (2009)
Lloyd et al., PRL 106, 040403 (2011)
Lloyd et al., PRD 84, 025007 (2011)

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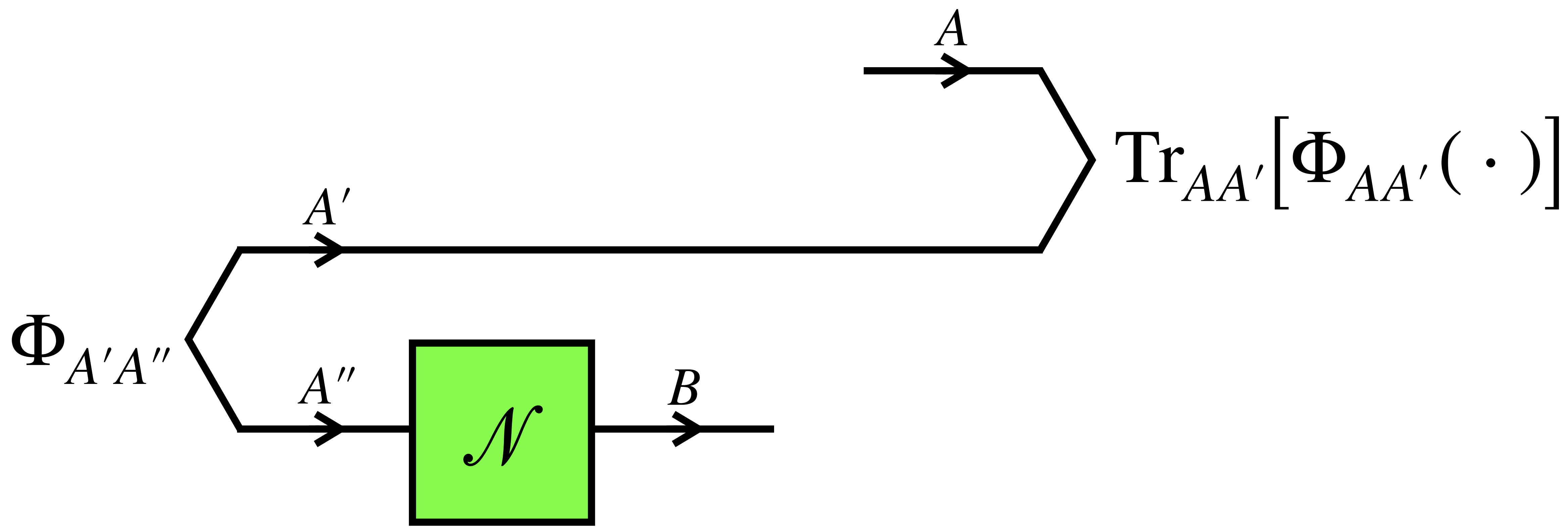

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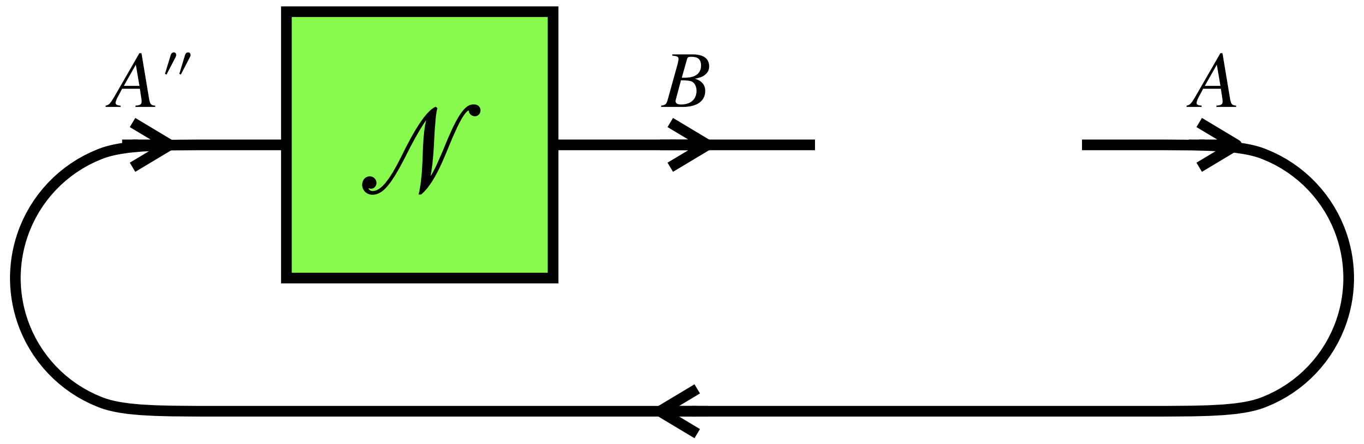

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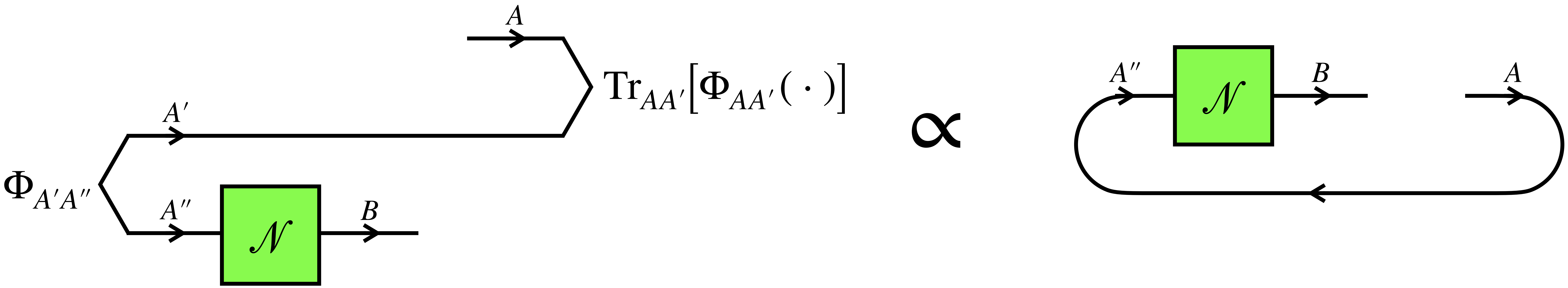
"Noisy" P-CTC



Genkina et al., PRA 85, 022330 (2012)

How to describe a noisy time machine mathematically?

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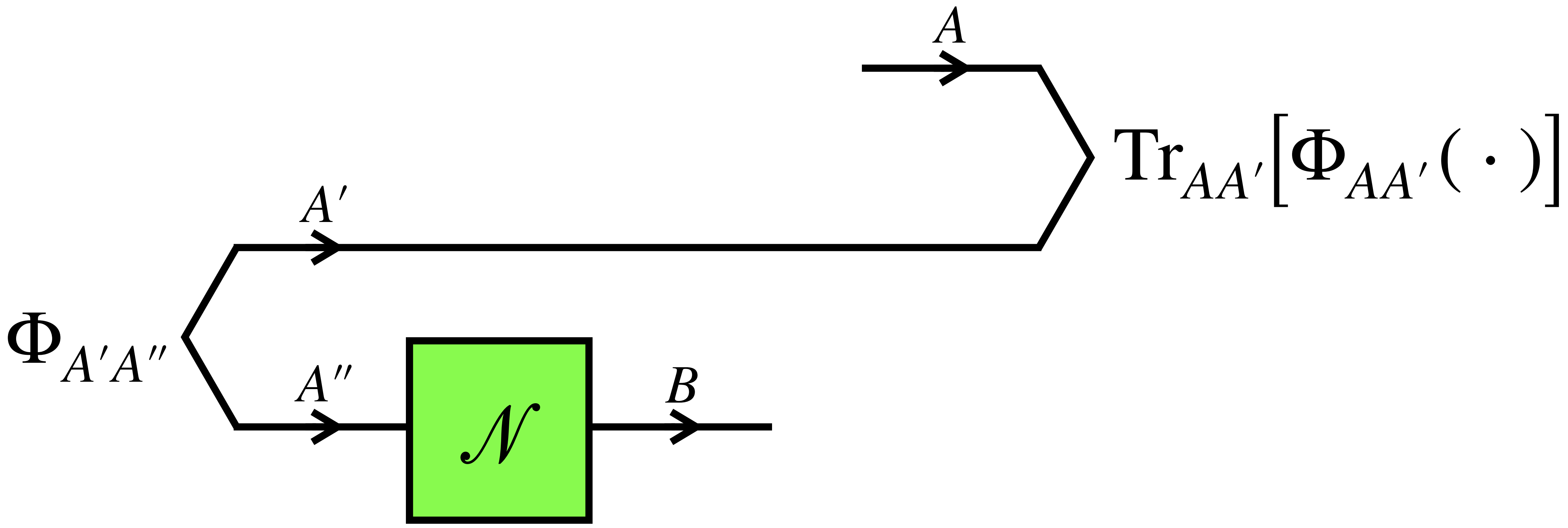
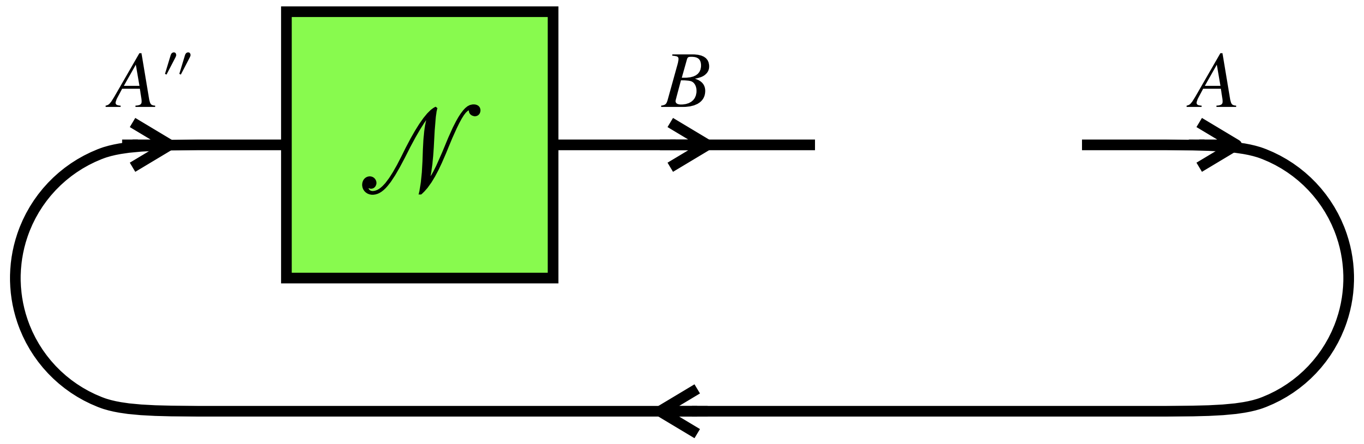


$\propto$



The effect of the noisy P-CTC represented by a CPTP map $\mathcal{N}_{A\to B}$:

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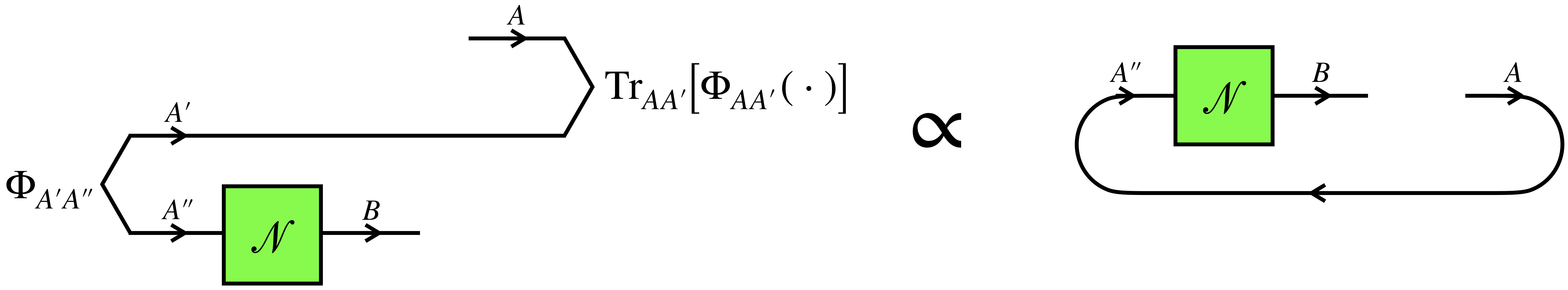

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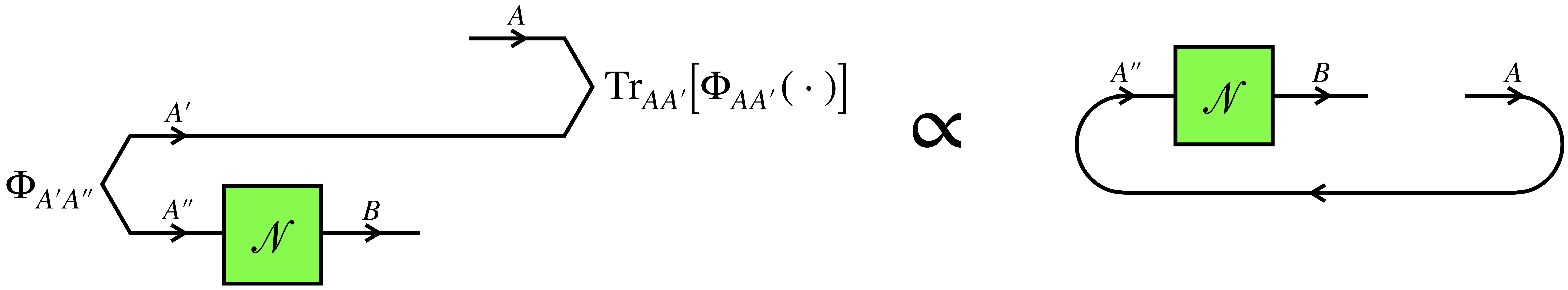

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- If we wait for B to "catch up", then it realizes an ordinary channel represented by $\mathcal{N}_{A\to B}$

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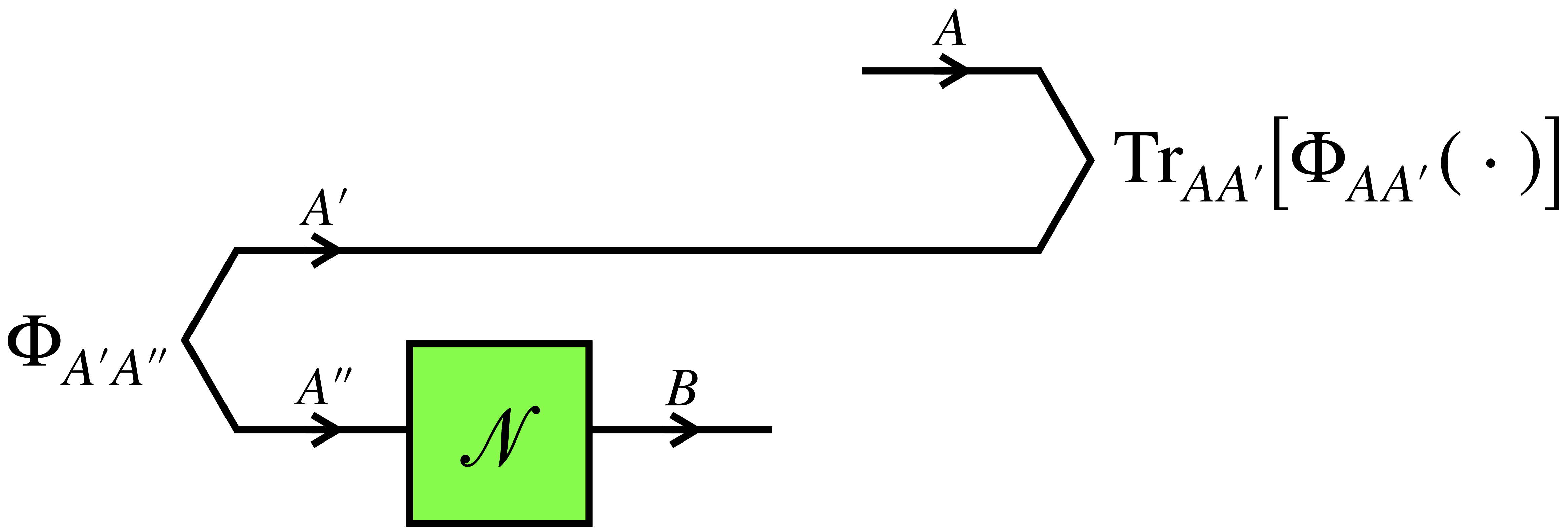

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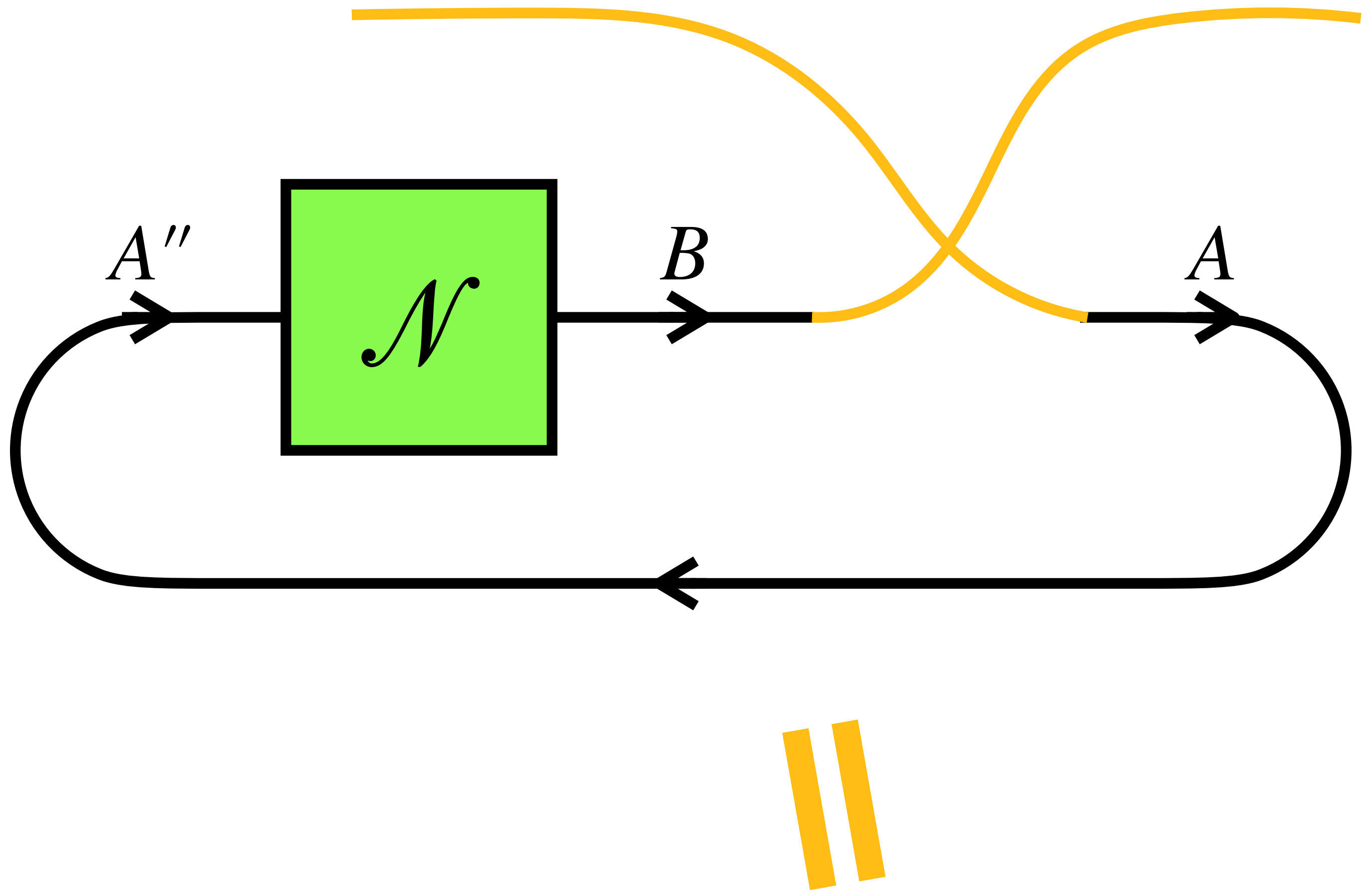

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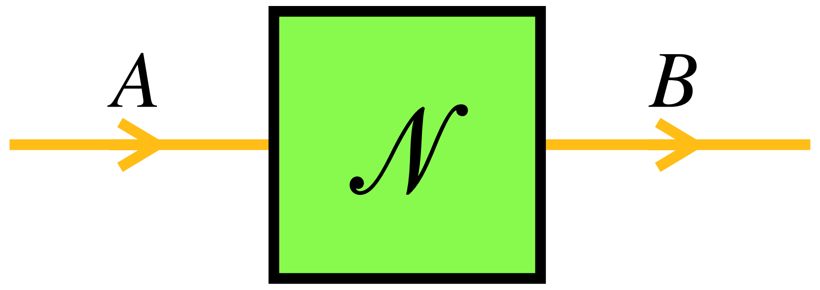

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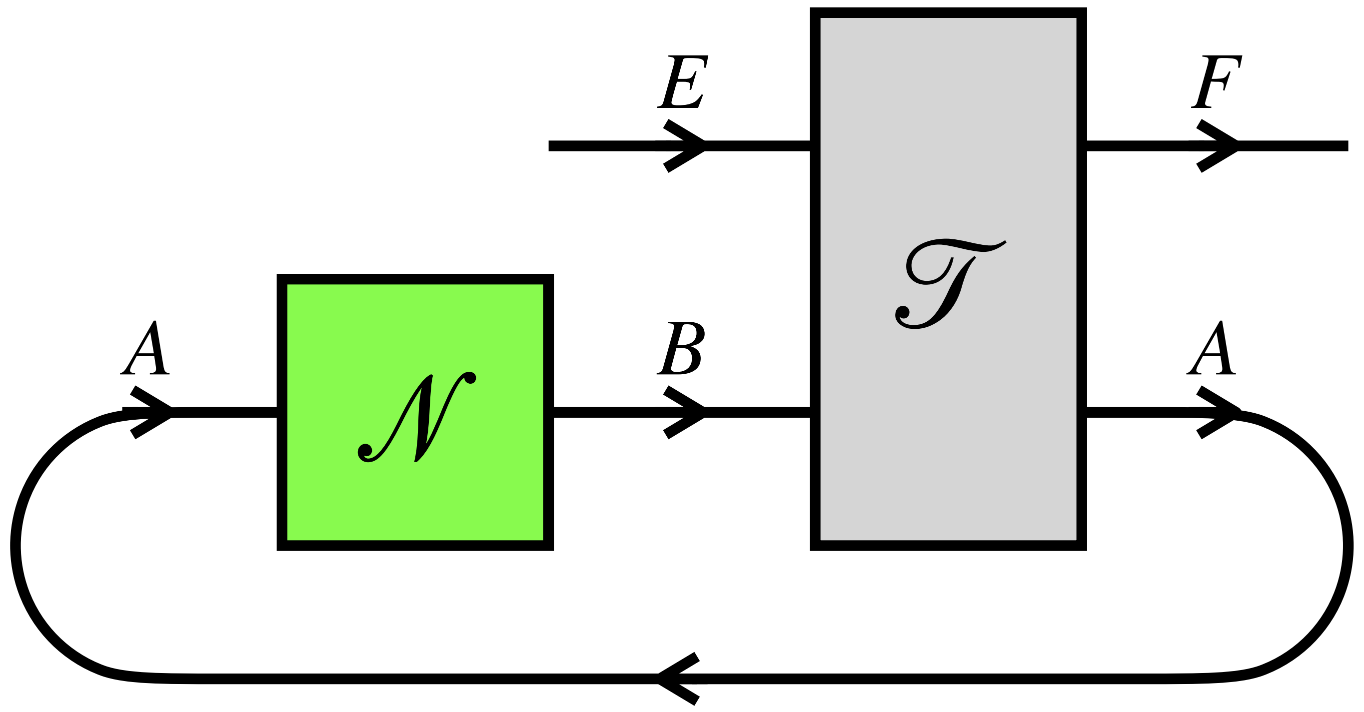
Combine a noisy P-CTC with an ordinary channel



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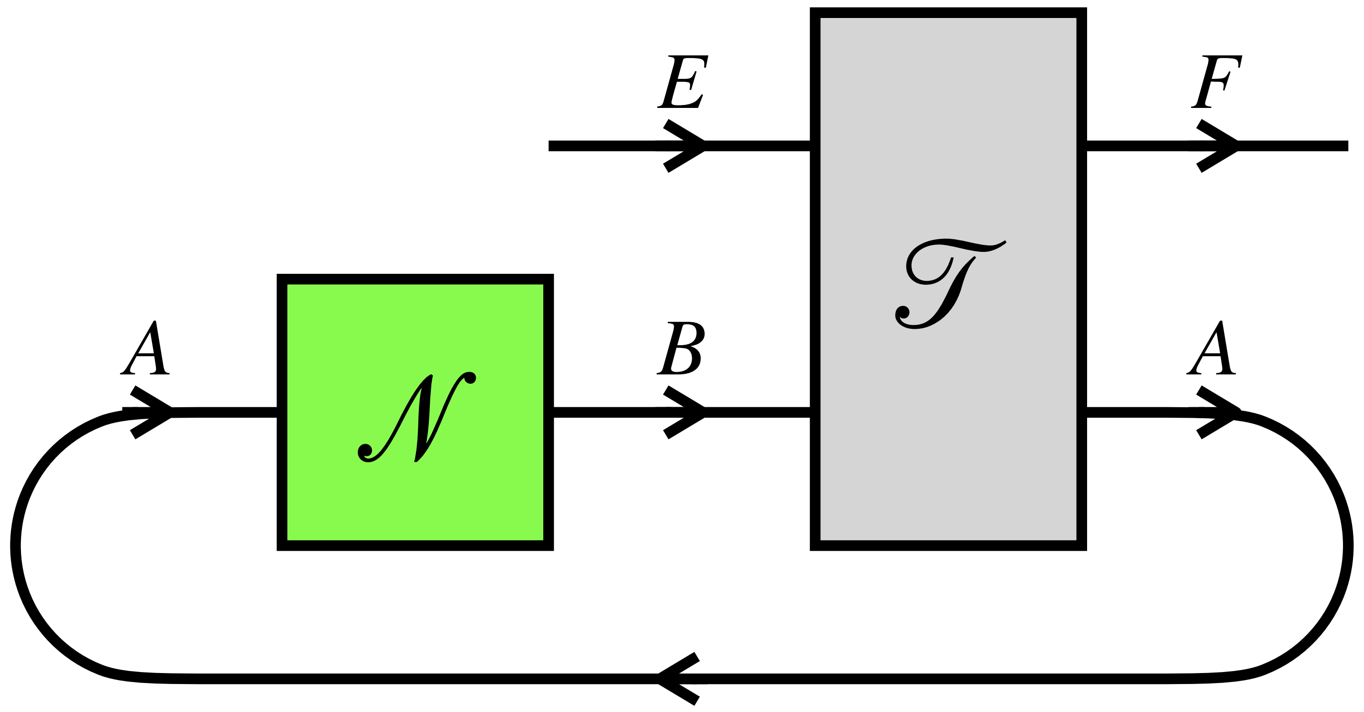

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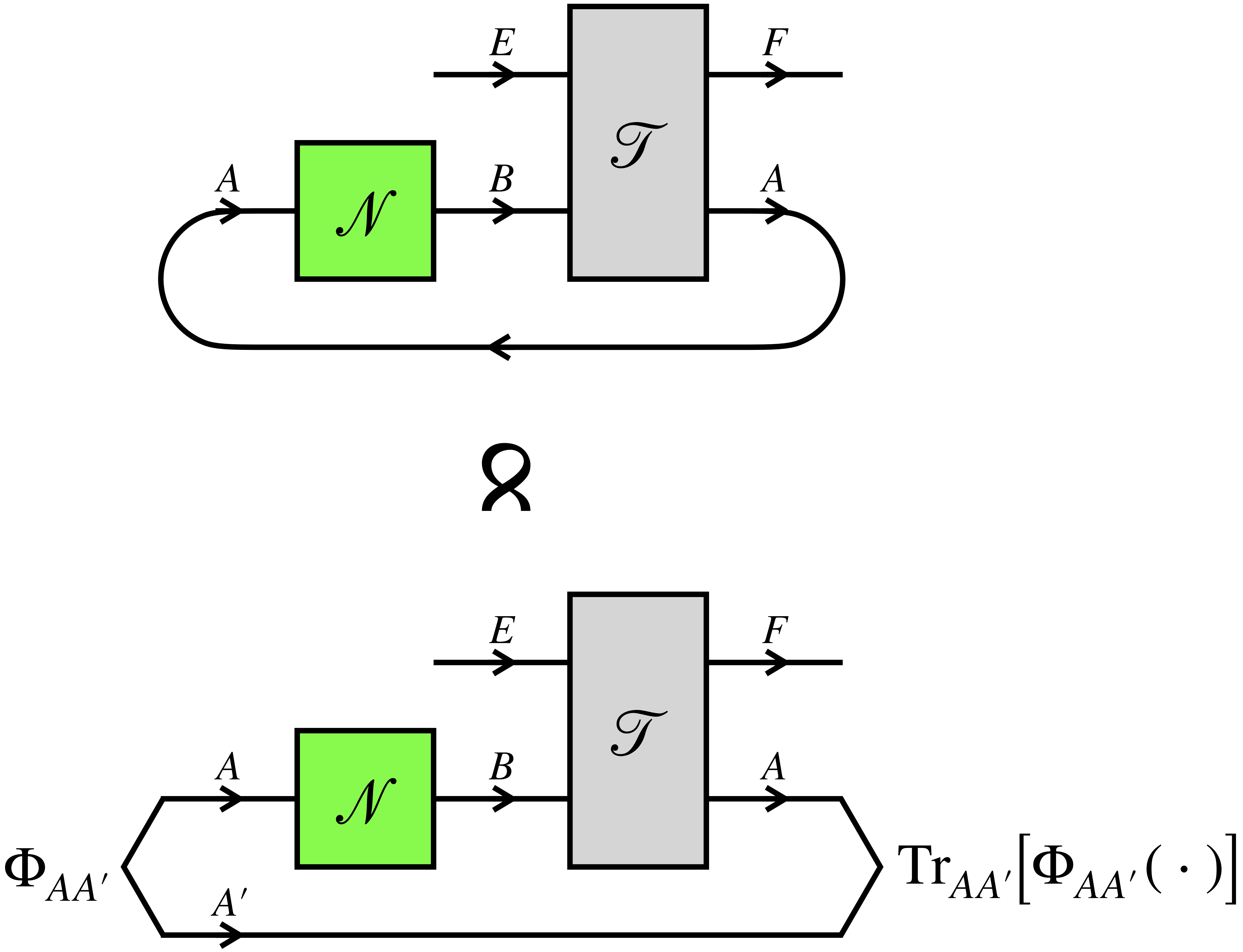

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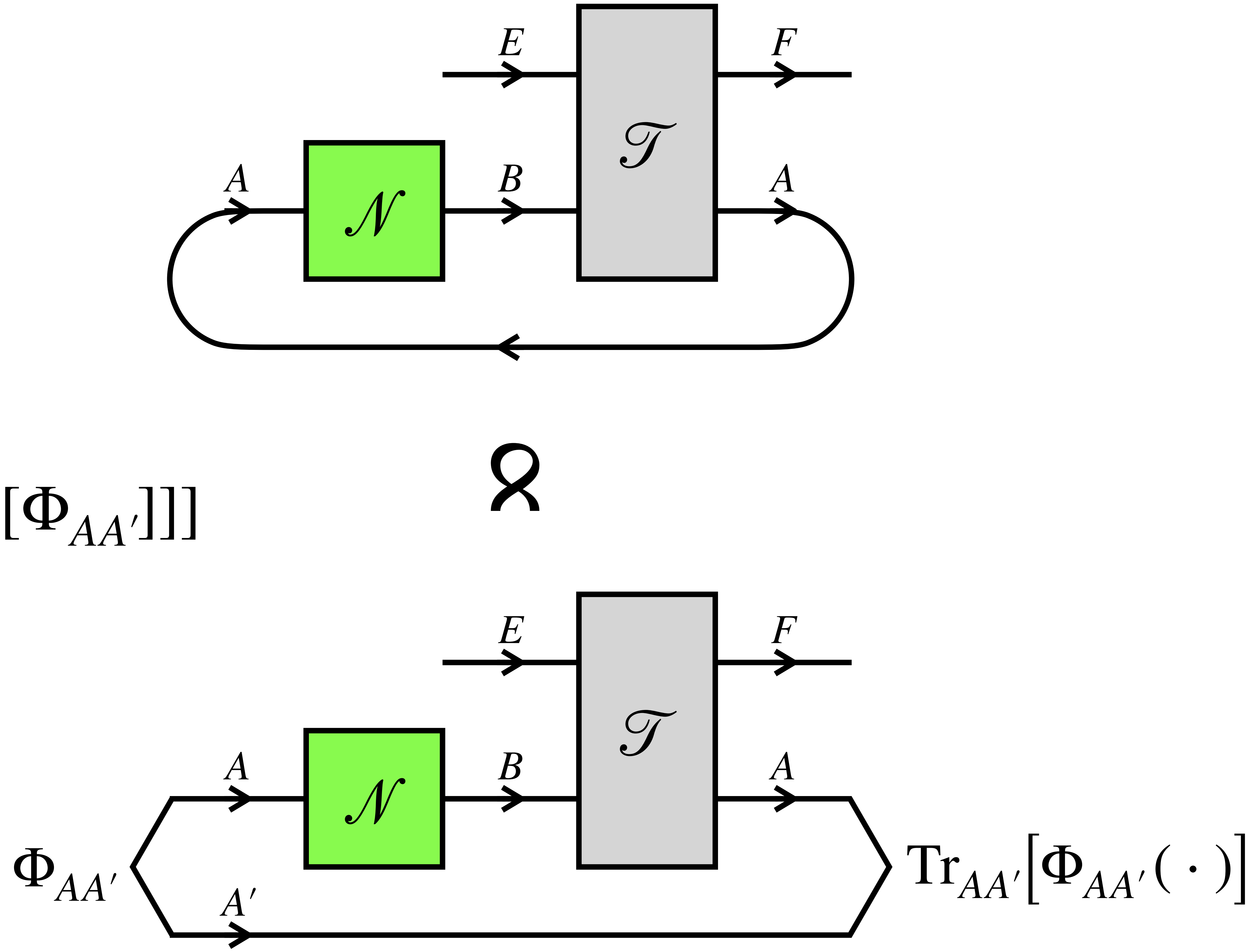
What is the map from E to F?

$$\rho_{RE} \mapsto \sigma_{RF} \propto \mathrm{Tr}_{AA'}[\Phi_{AA'}\mathcal{T}_{EB\to FA}[\rho_{RE} \otimes \mathcal{N}_{A\to B}[\Phi_{AA'}]]]$$

- nonlinear due to renormalization



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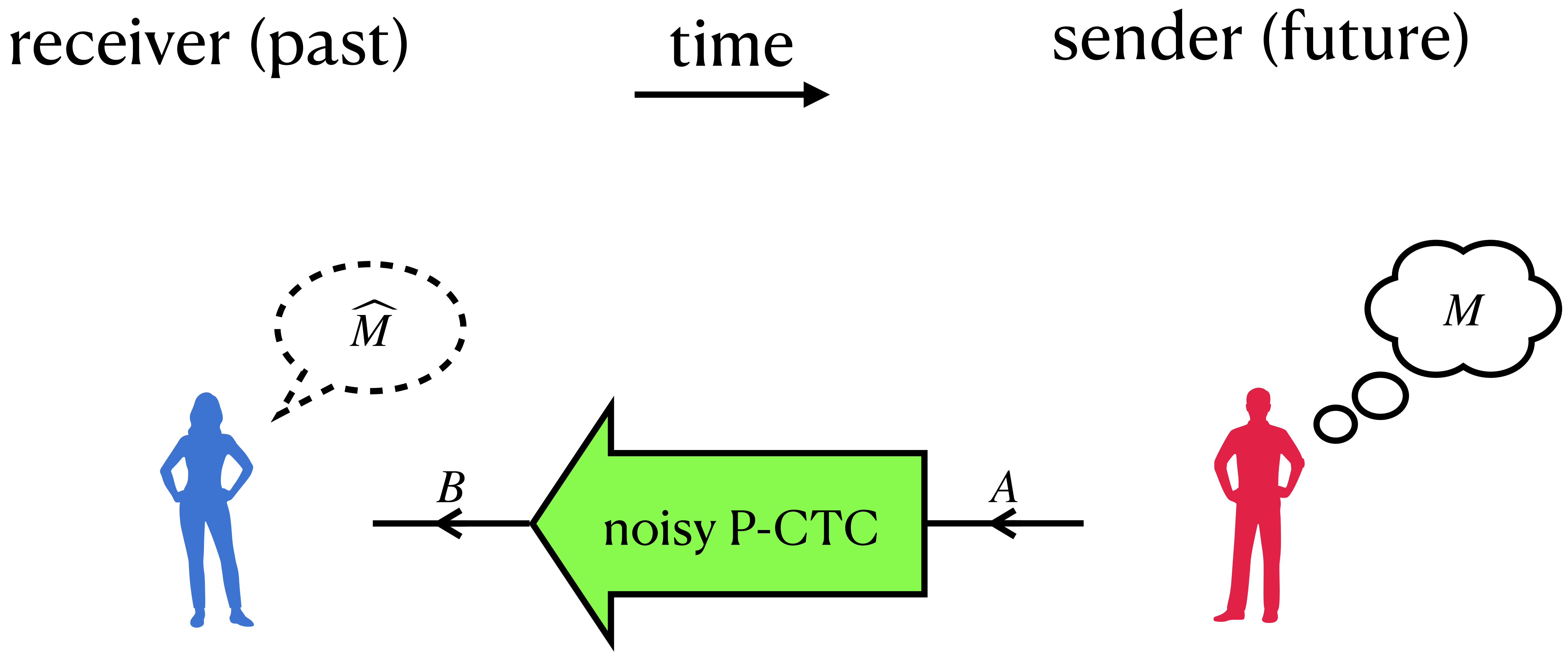
“Retrocausal” communication:

Messaging the past from the future

Retrocausal communication

Setting: a sender in the future is connected to a receiver in the past through a noisy P-CTC

Goal: transmit quantum or classical messages reliably from the sender to the receiver, as efficiently as possible, using whatever strategy allowed by physics



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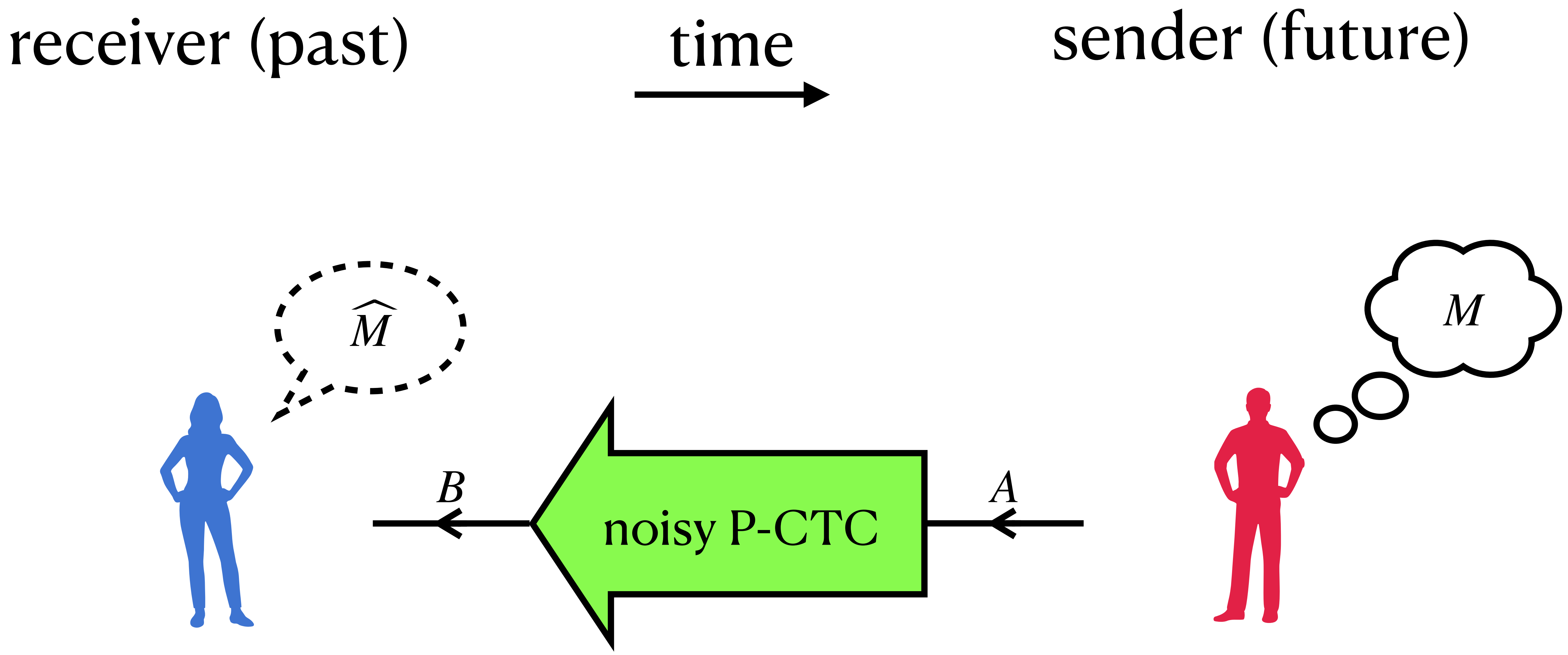

Retrocausal communication

father

Setting: a sender in the future is connected to a receiver in the past through a noisy P-CTC

daughter

bookshelf

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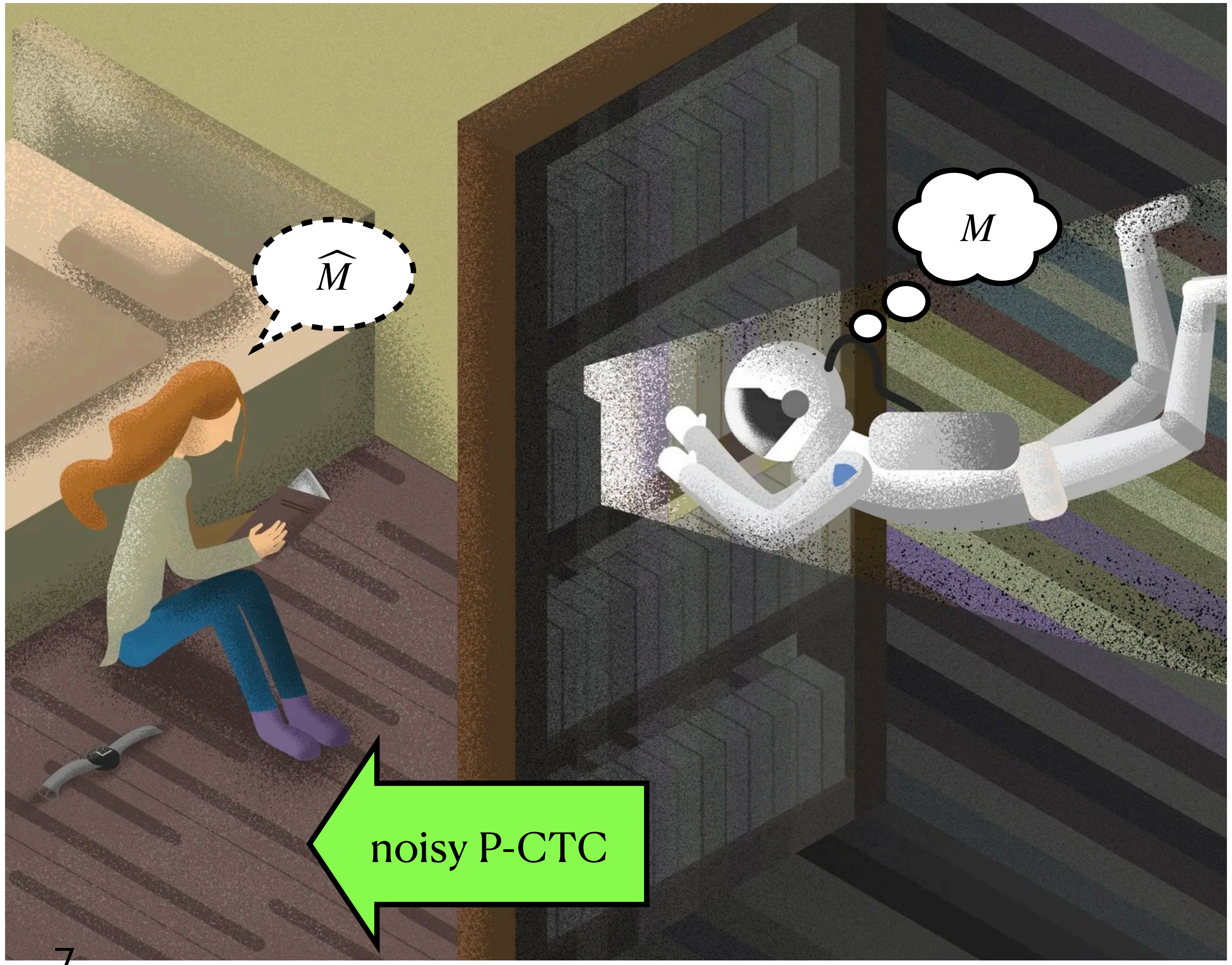
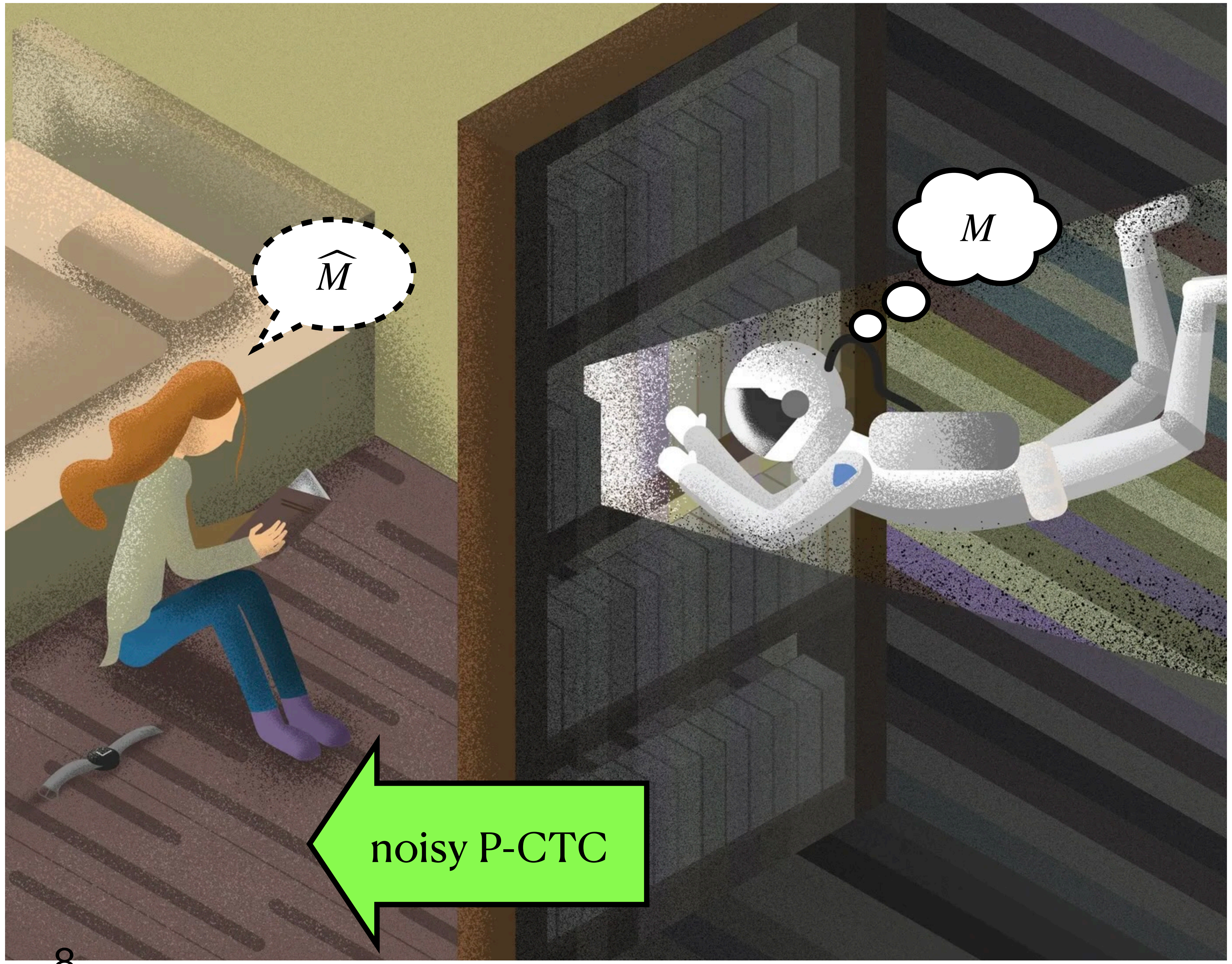
receiver (past) time sender (future)



Bookshelf Scene (2018), ©Alexey Nefed (motion.nalex) receiver (past) time sender (future)



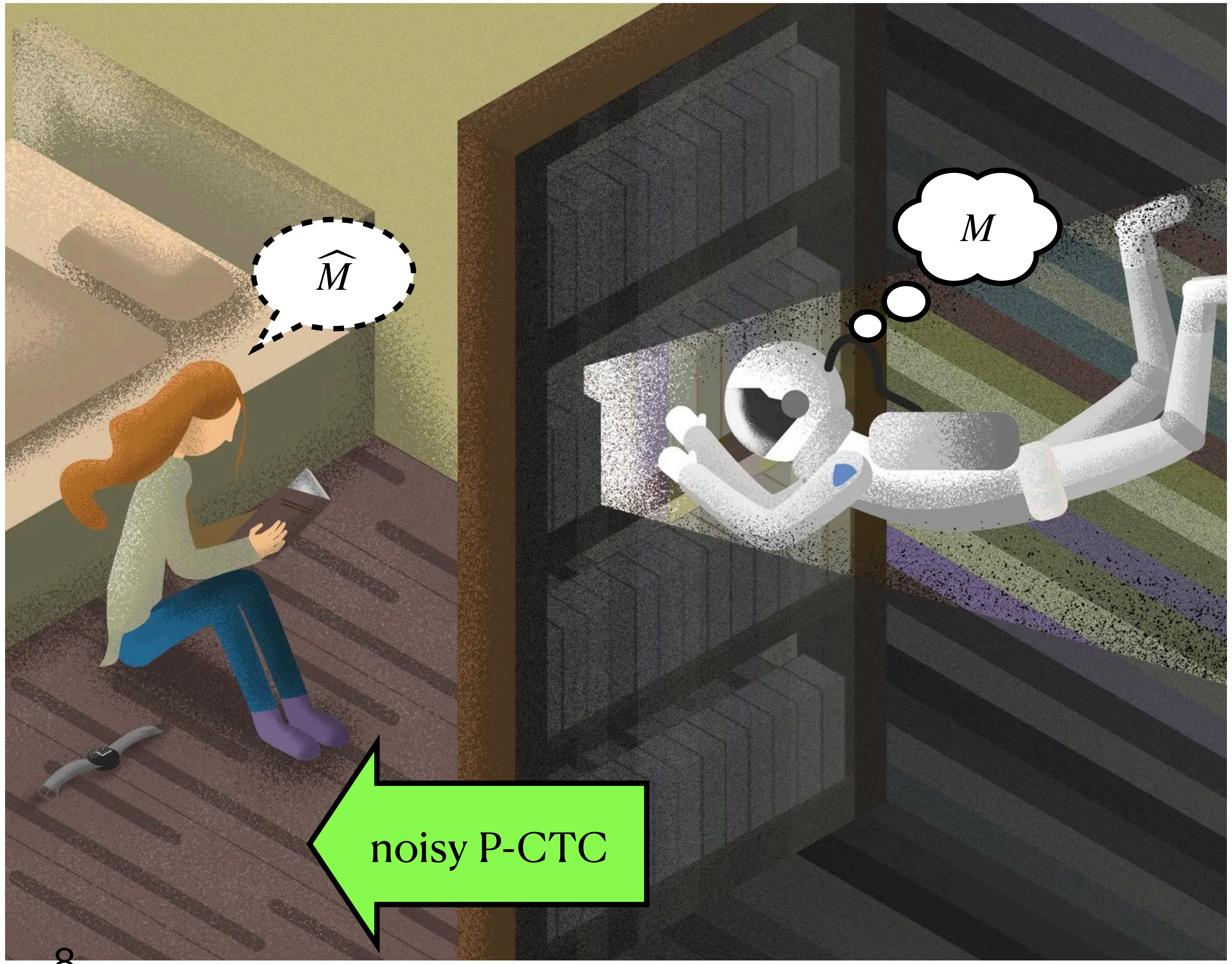
General strategy for retrocausal communication

Strategy:

receiver (past)

time

sender (future)



General strategy for retrocausal communication

Strategy:

- Encoding
- Decoding

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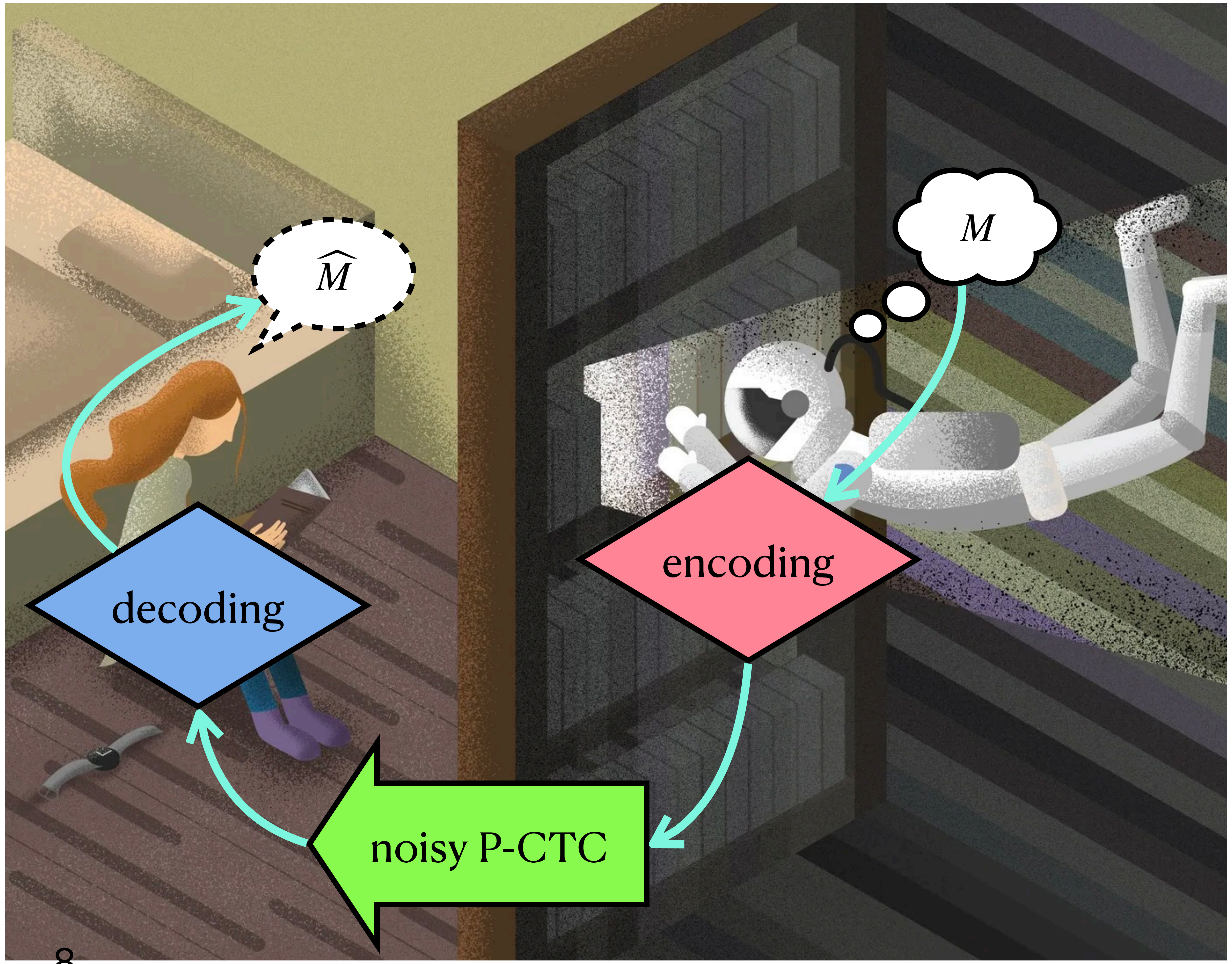

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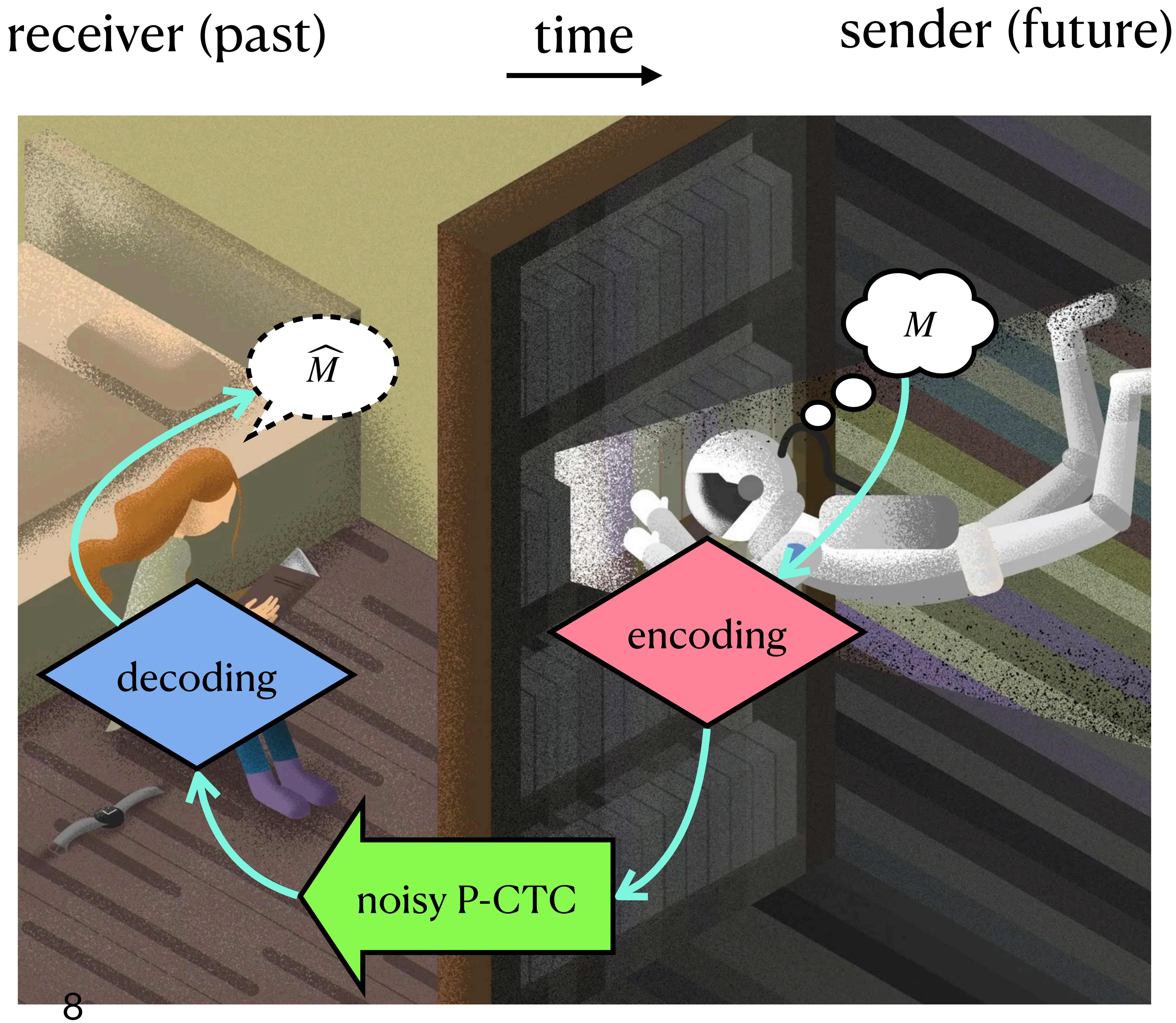

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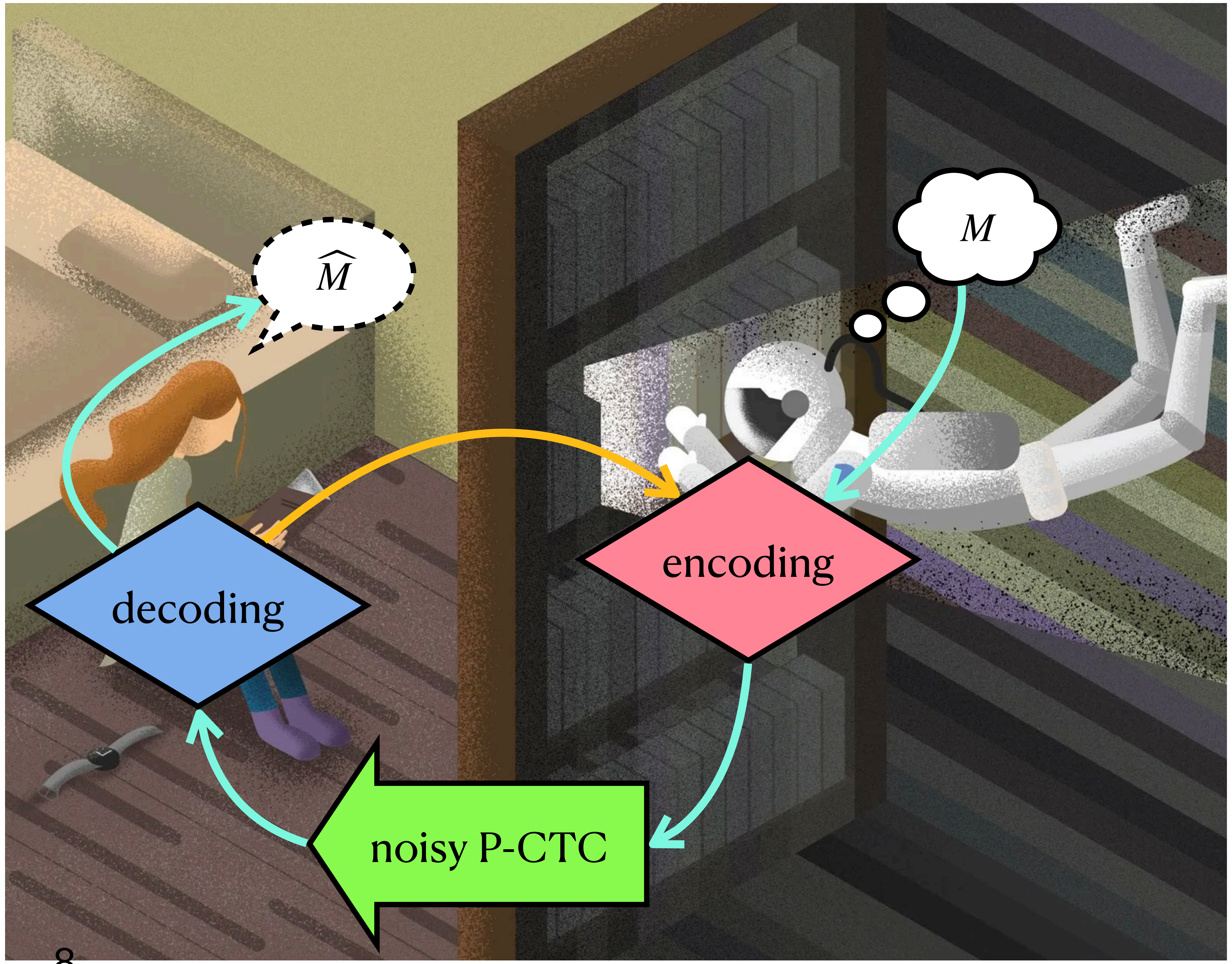

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- Causal loop, but perfectly reasonable

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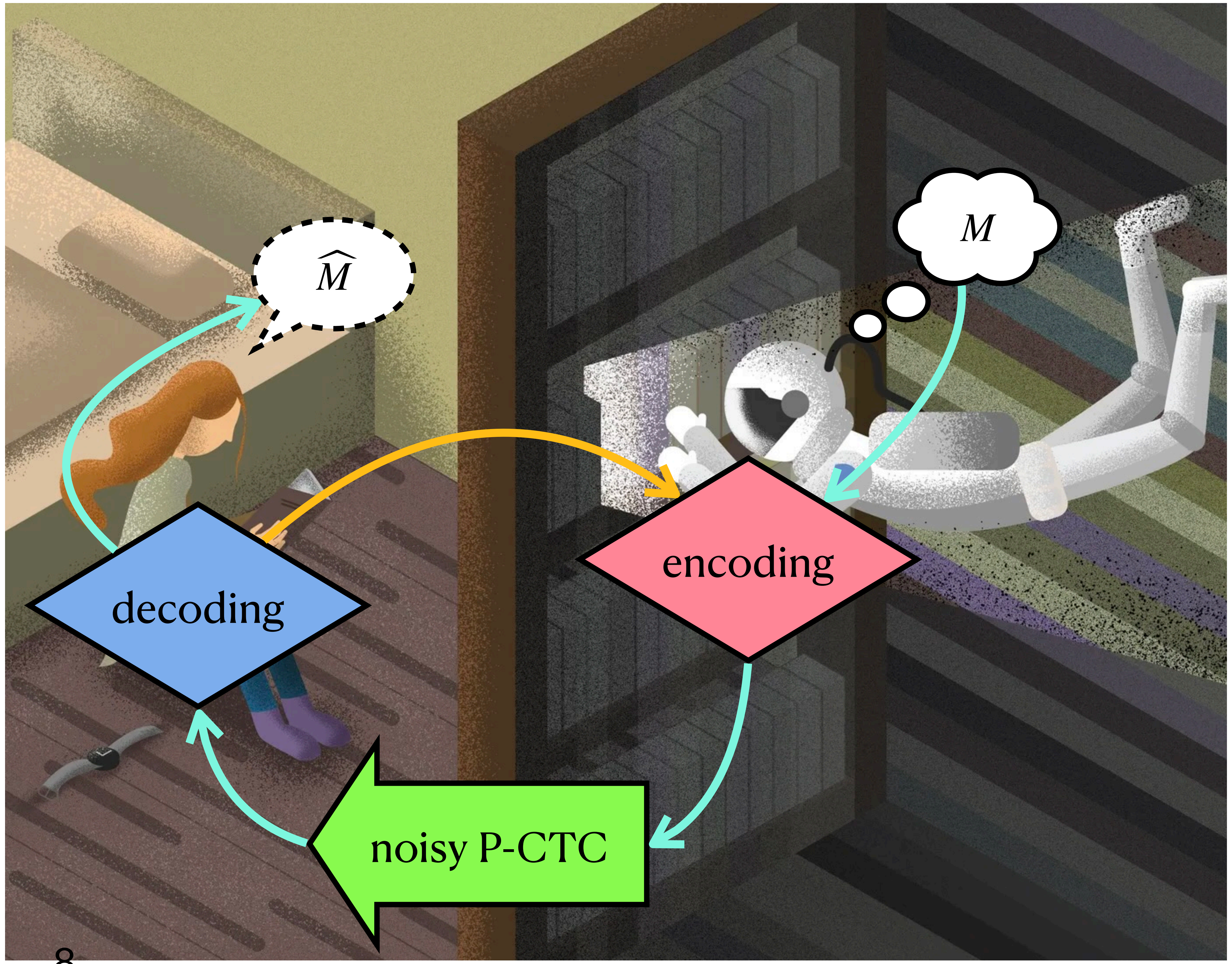

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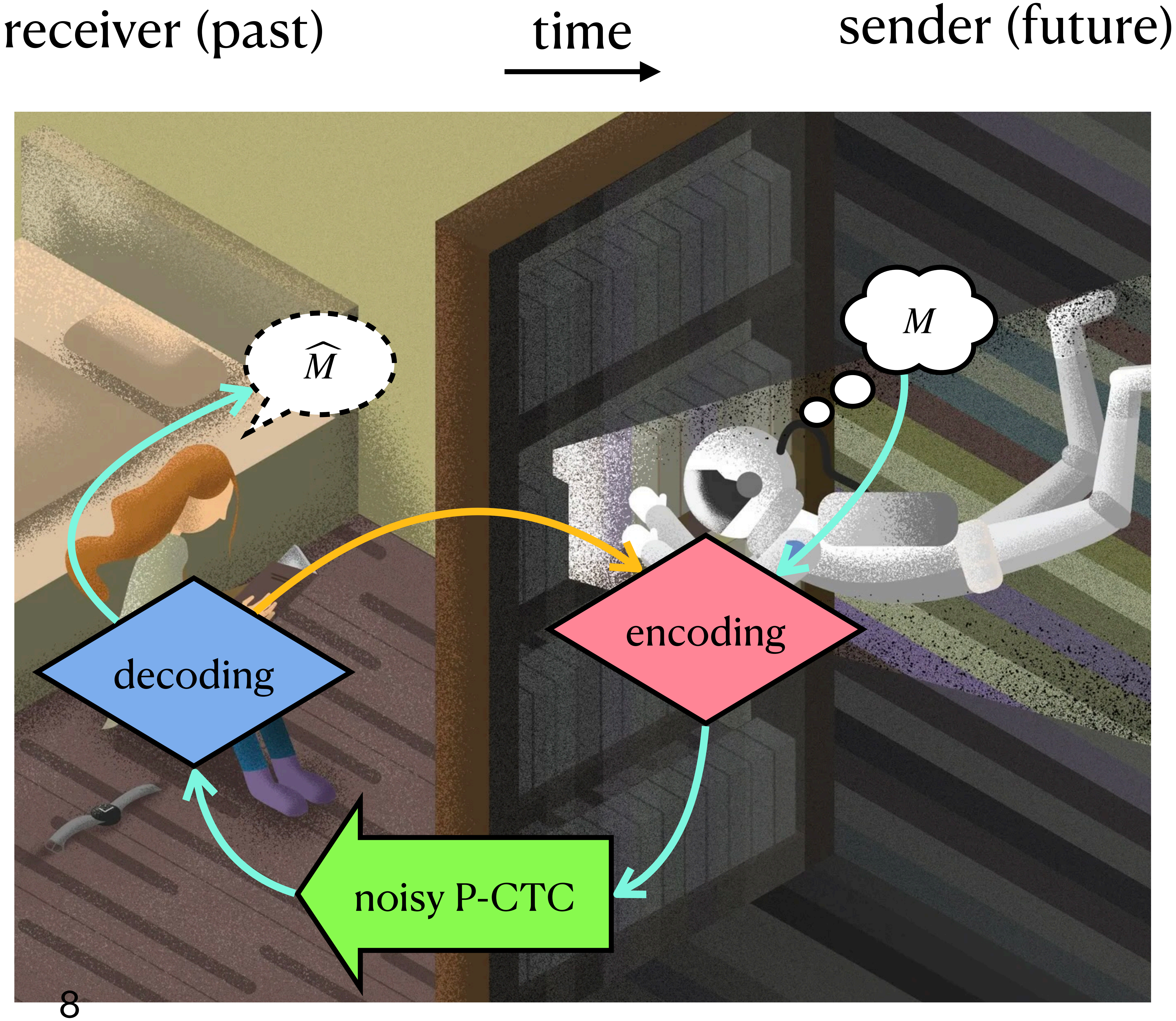

General strategy for retrocausal communication

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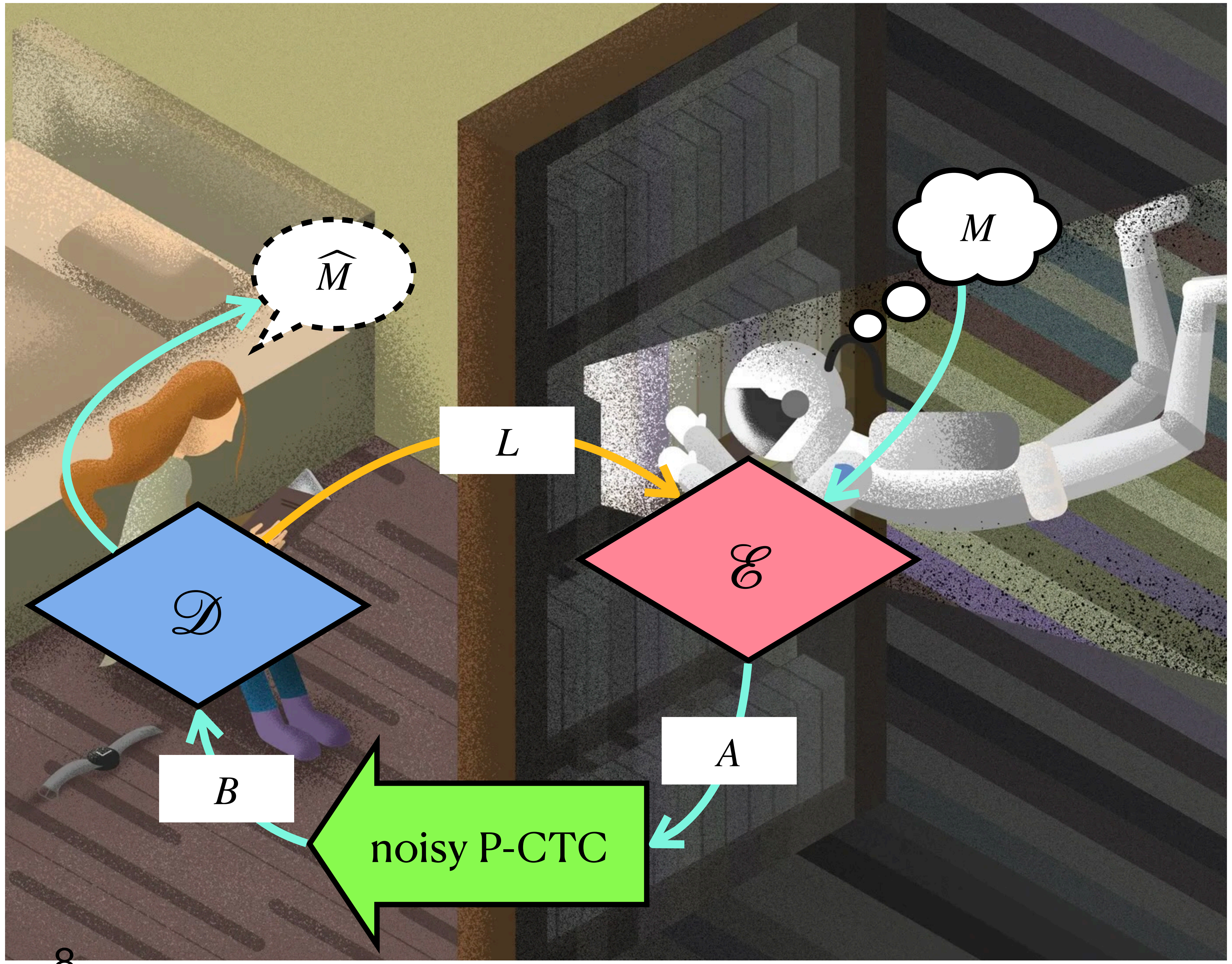

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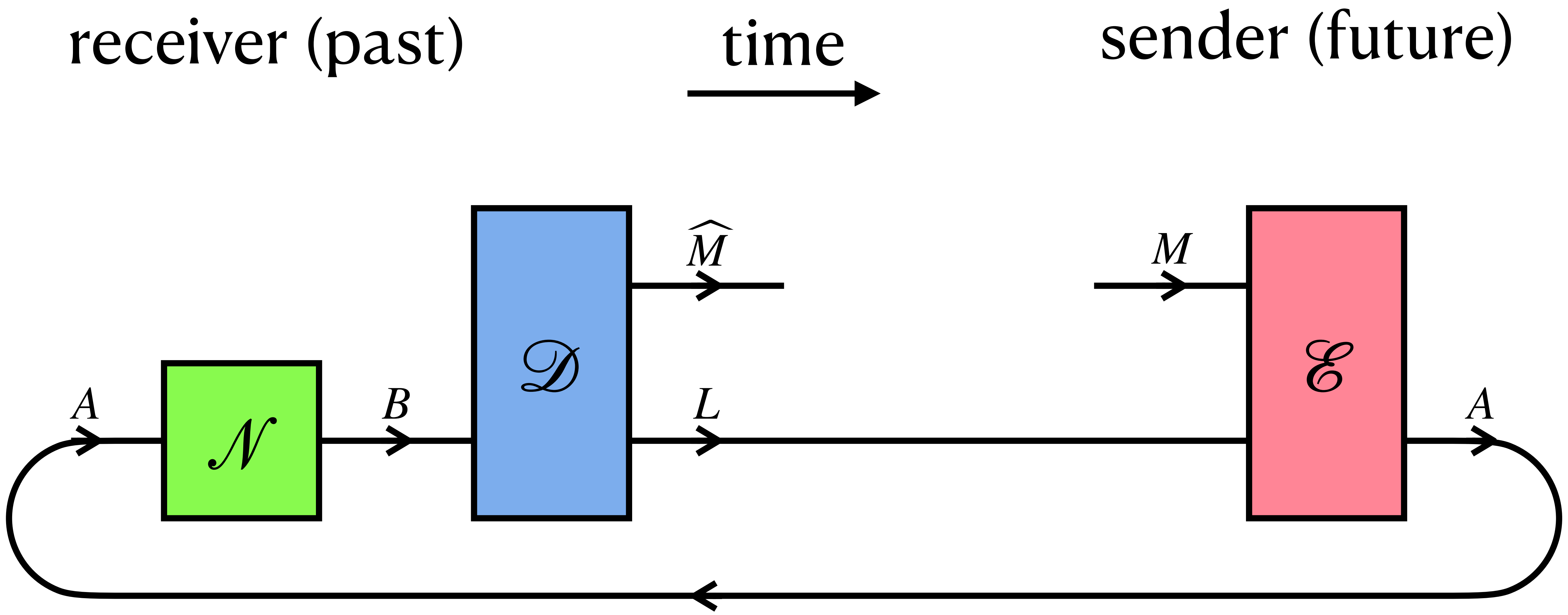

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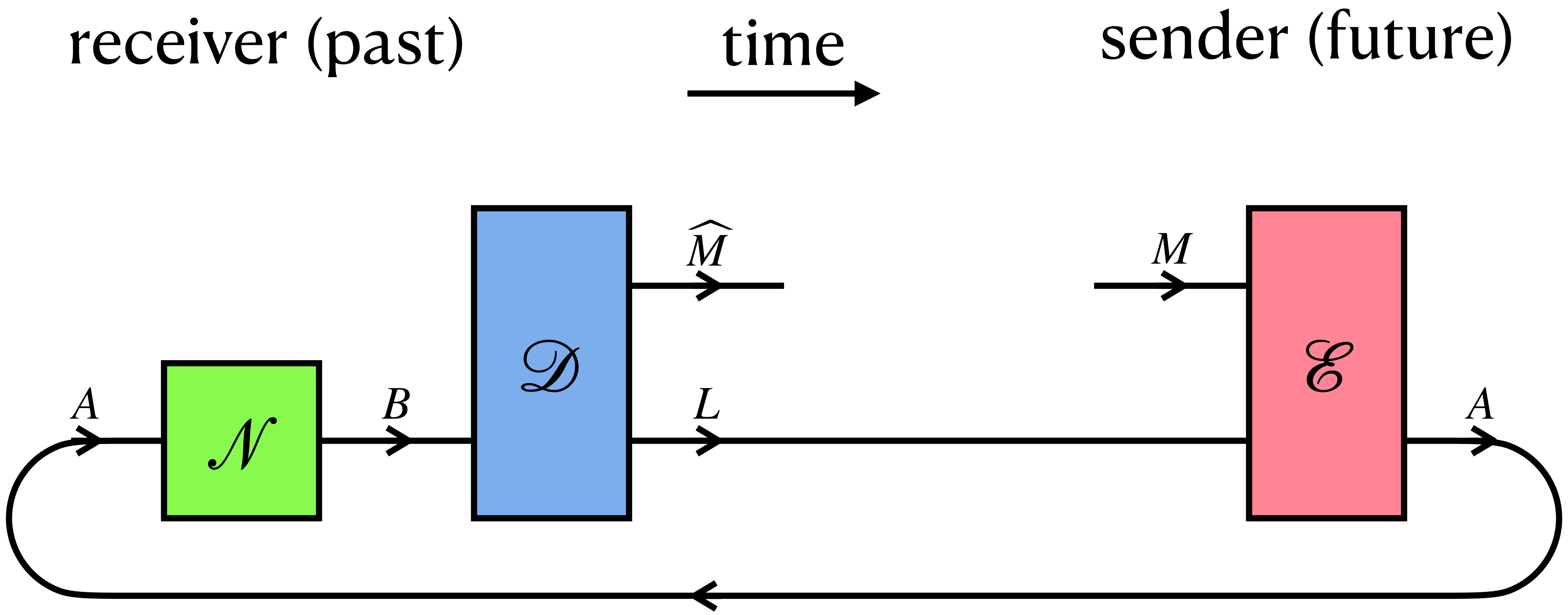
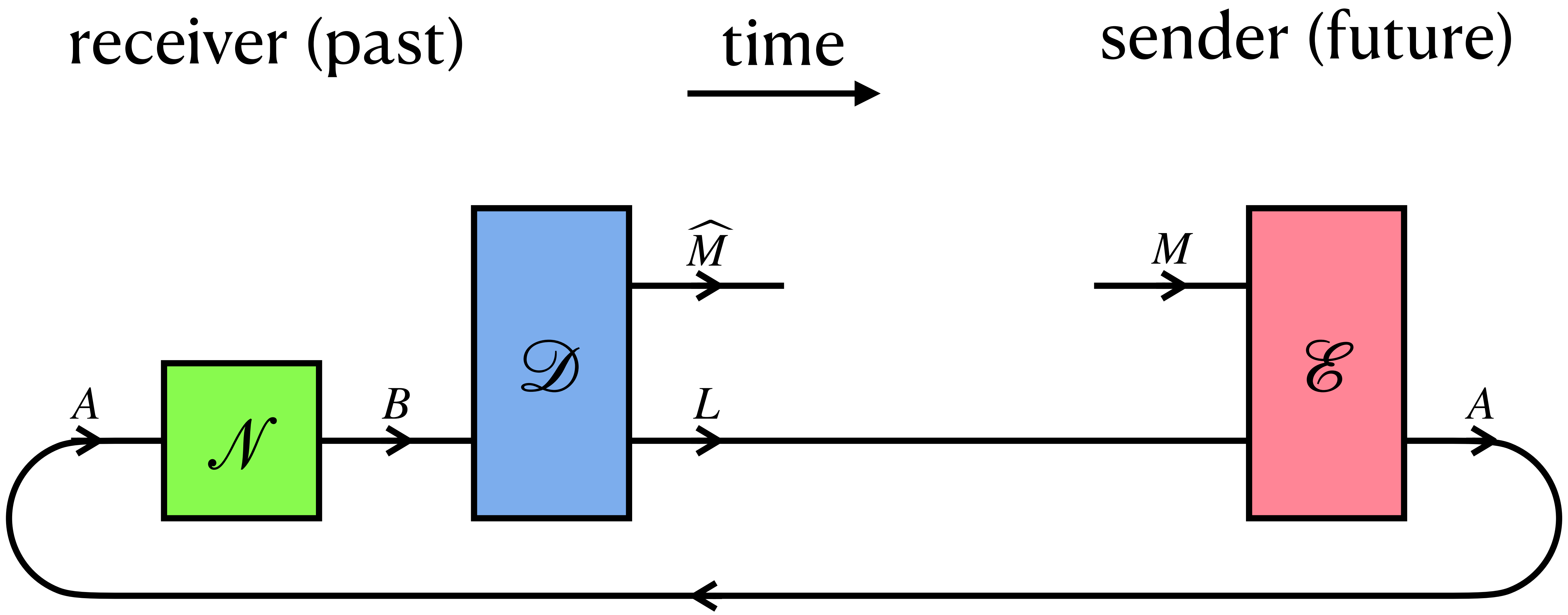


Call the entire map $\mathscr{M}_{M \to \widehat{M}} \equiv \mathscr{M}^{(\mathscr{N},\mathscr{E},\mathscr{D})}_{M \to \widehat{M}}$



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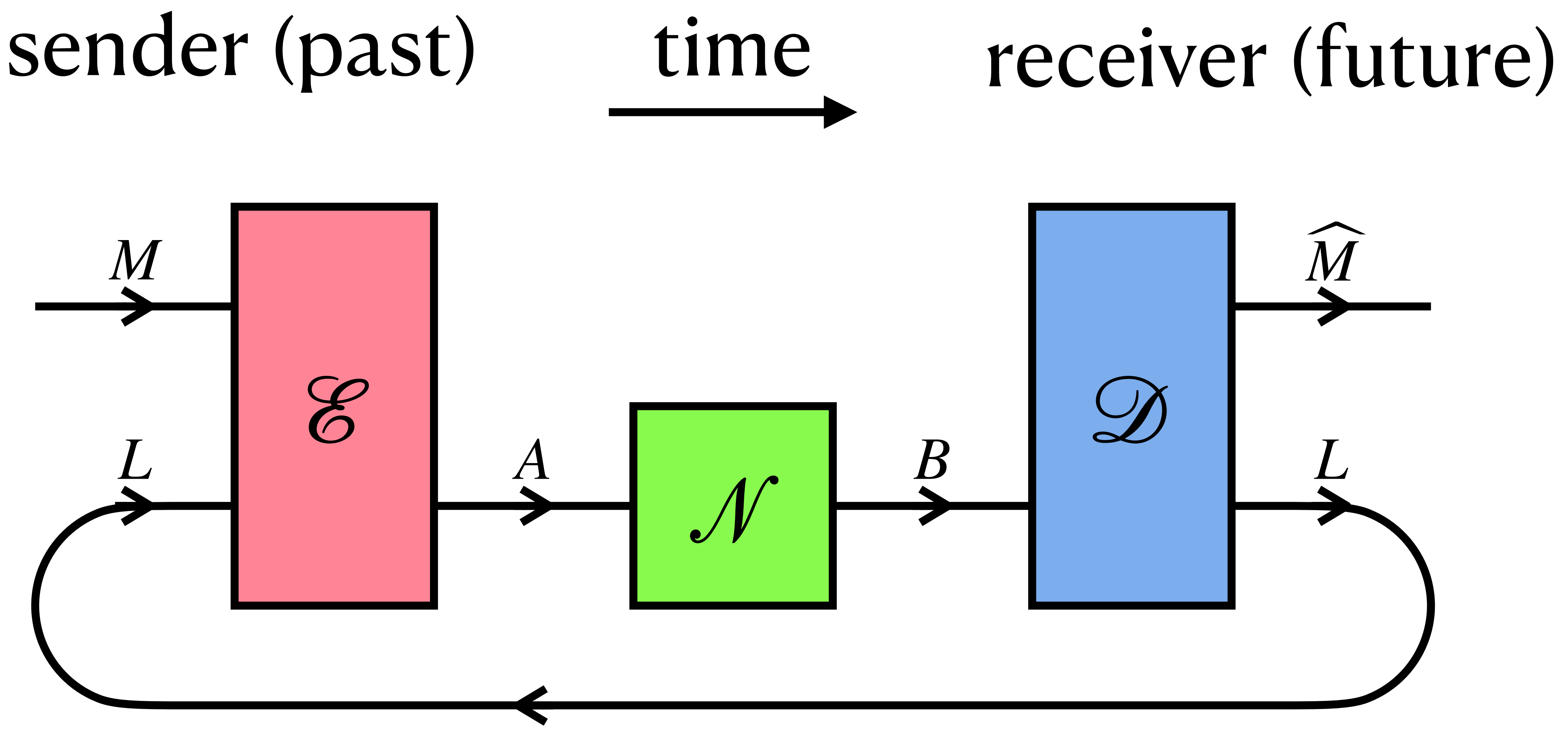

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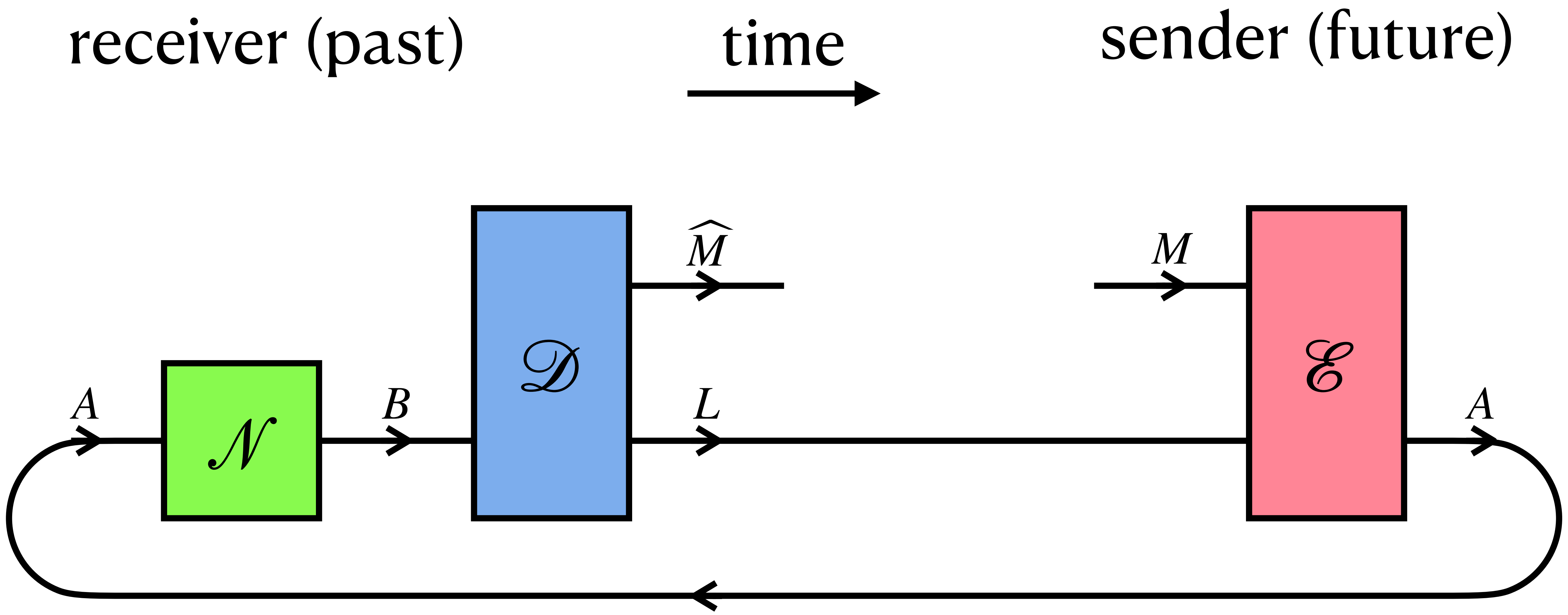

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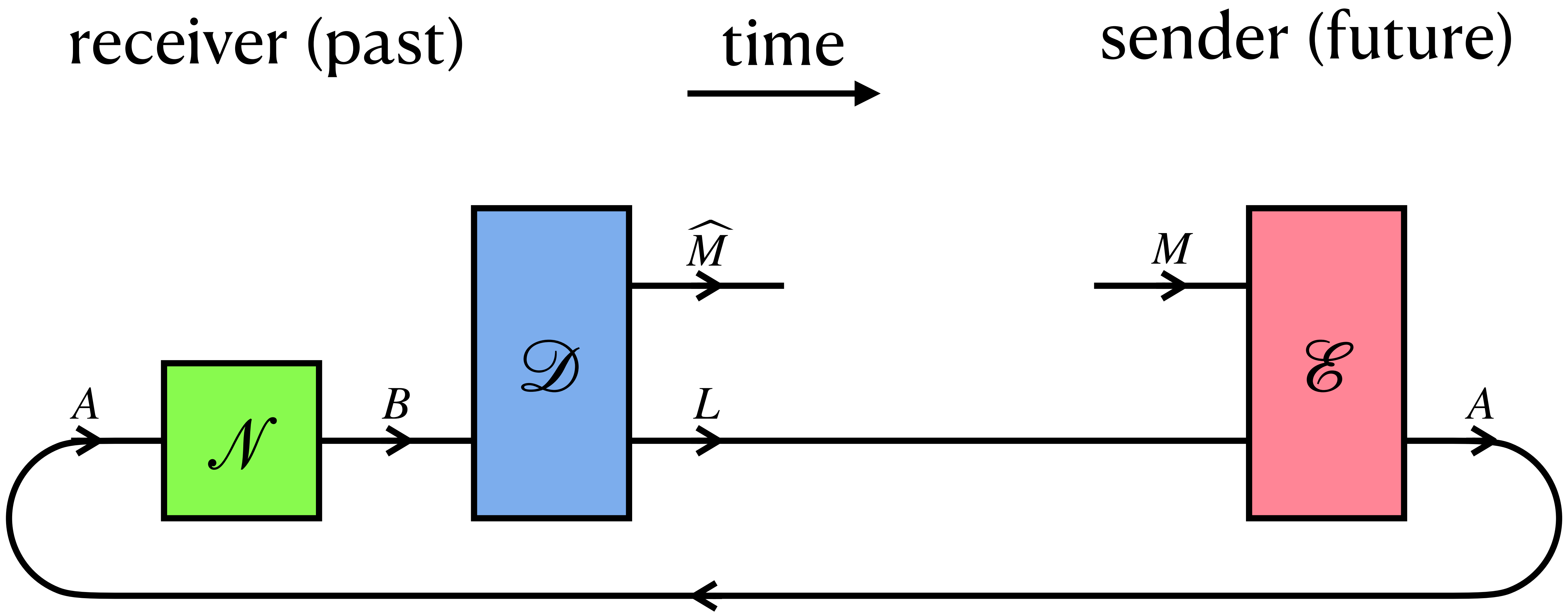

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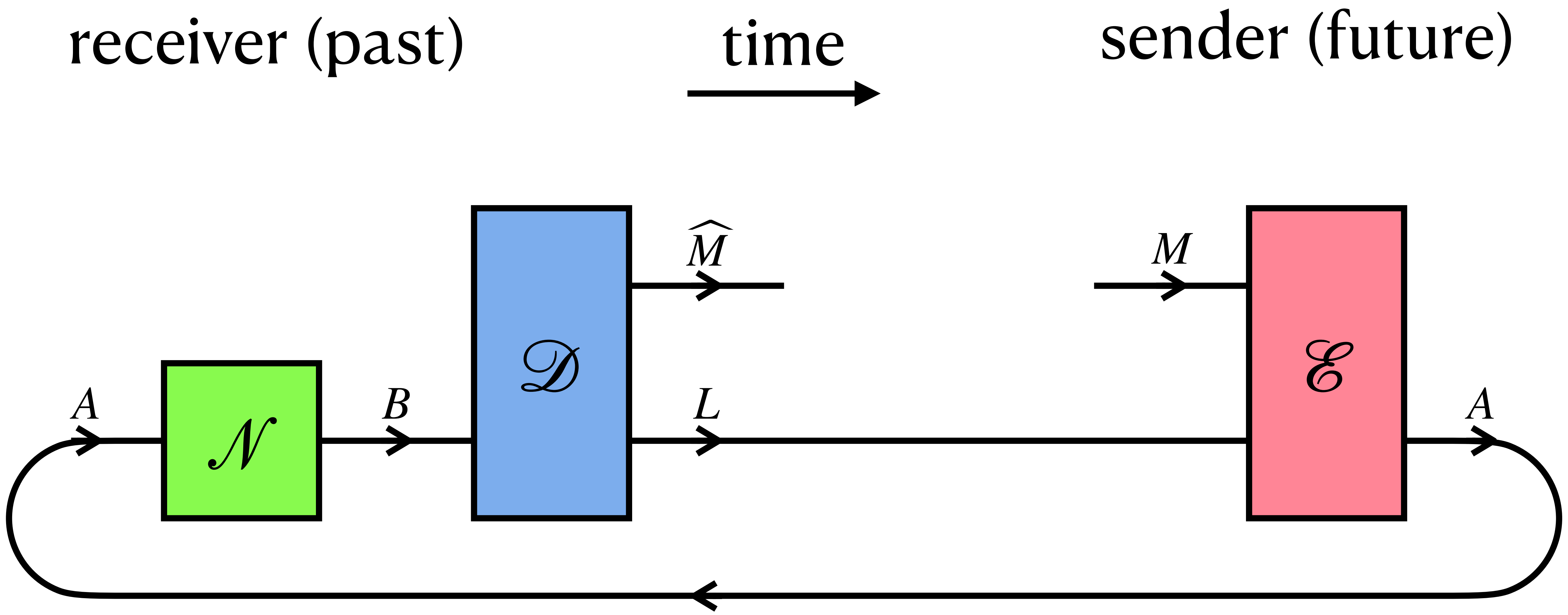
Definition: One-shot retrocausal quantum & classical capacity



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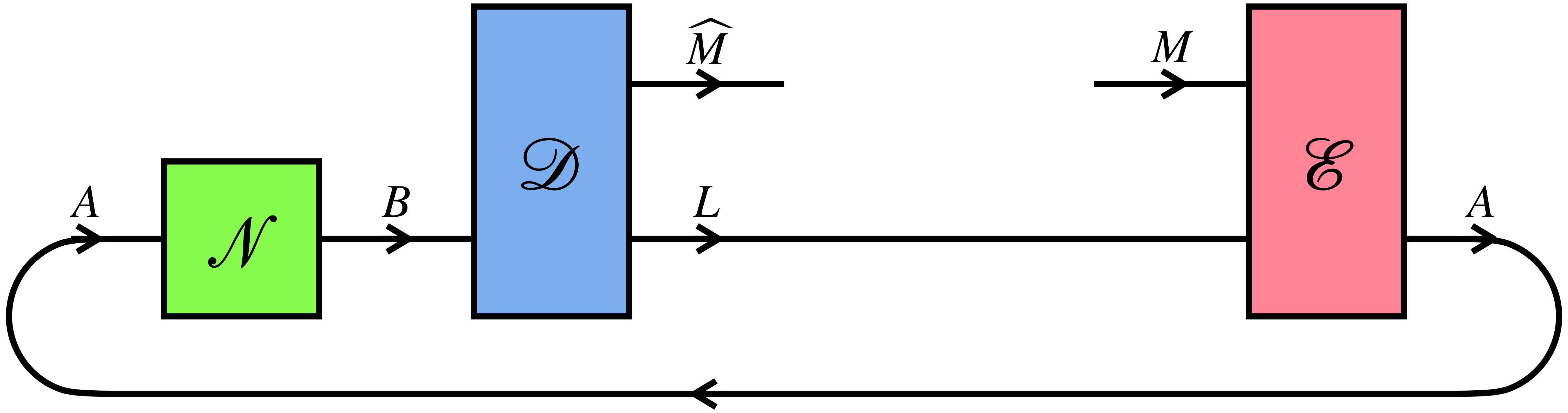
Definition: One-shot retrocausal quantum & classical capacity

One-shot retrocausal quantum capacity:

$$Q^{\varepsilon}_{\text{retro}}(\mathcal{N}) := \sup_{\mathcal{E}_{ML\to A}, \mathcal{D}_{B\to \widehat{M}L}} \log_2 d_M$$

$$\text{s.t.} \ \inf_{\psi_{RM}} \mathrm{Tr}[\psi_{R\widehat{M}} \mathcal{M}_{M\to\widehat{M}}[\psi_{RM}]] \geq 1-\varepsilon$$

receiver (past) time sender (future)



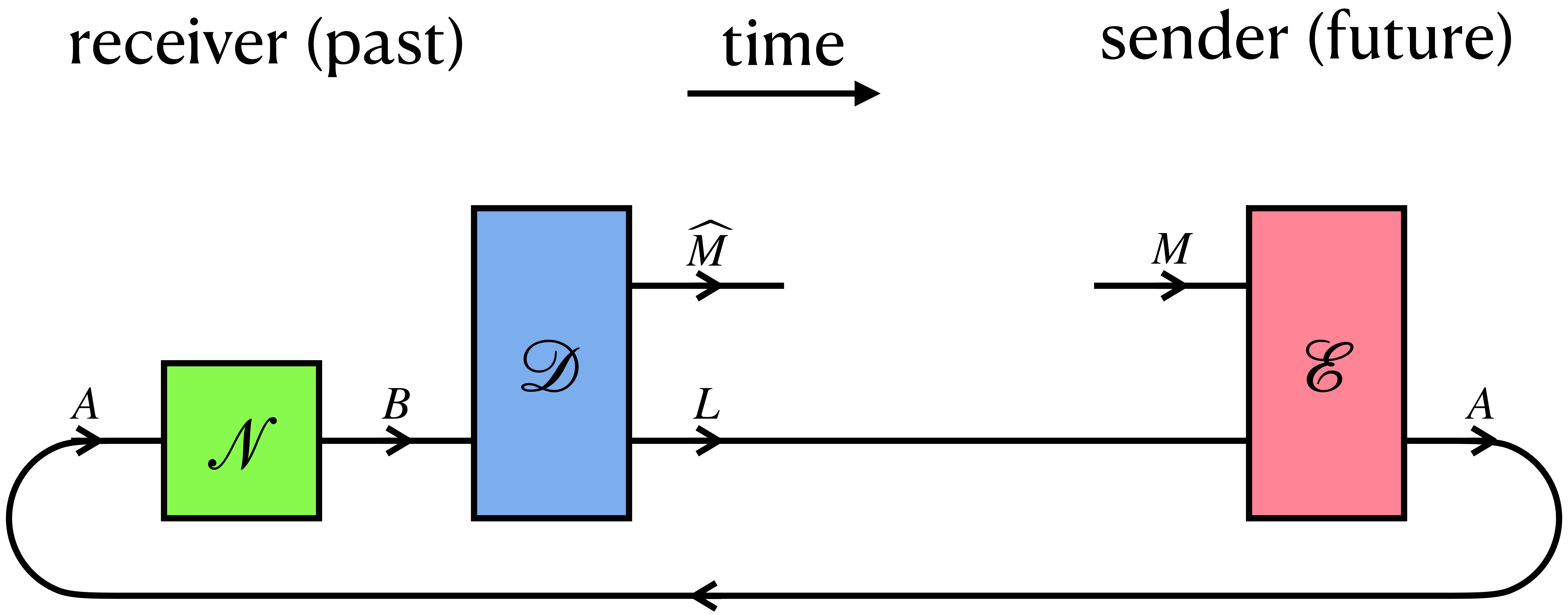
Call the entire map $\mathcal{M}_{M\to\widehat{M}} \equiv \mathcal{M}^{(\mathcal{N},\mathcal{E},\mathcal{D})}_{M\to\widehat{M}}$

Definition: One-shot retrocausal quantum & classical capacity

One-shot retrocausal quantum capacity:

$$Q^{\varepsilon}_{\text{retro}}(\mathcal{N}) := \sup_{\mathcal{E}_{ML\to A}, \mathcal{D}_{B\to \widehat{M}L}} \log_2 d_M$$

$$\text{s.t. } \inf_{\psi_{RM}} \text{Tr}[\psi_{R\widehat{M}} \mathcal{M}_{M\to\widehat{M}}[\psi_{RM}]] \geq 1 - \varepsilon$$



One-shot retrocausal classical capacity:

Call the entire map $\mathcal{M}_{M\to\widehat{M}} \equiv \mathcal{M}^{(\mathcal{N},\mathcal{E},\mathcal{D})}_{M\to\widehat{M}}$

$$C^{\varepsilon}_{\text{retro}}(\mathcal{N}) := \sup_{\mathcal{E}_{ML\to A}, \mathcal{D}_{B\to \widehat{M}L}} \log_2 d_M$$

$$\text{s.t. } \min_{1\leq m\leq d_M} \text{Tr}[|m\rangle\langle m|_{\widehat{M}} \mathcal{M}_{M\to\widehat{M}}[|m\rangle\langle m|_M]] \geq 1 - \varepsilon$$

Main result: Noisy-"time-machine" coding theorem

One-shot retrocausal quantum & classical capacity: for all $\varepsilon \in [0,1]$,

$$Q^{\varepsilon}_{\mathrm{retro}}(\mathcal{N}) = \log_2 \left\lfloor \sqrt{\frac{\varepsilon}{1-\varepsilon} 2^{I_{\max}(\mathcal{N}) + I_{\mathrm{doe}}(\mathcal{N})} + 1} \right\rfloor$$

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Asymptotic retrocausal quantum & classical capacity: for all $\varepsilon \in (0,1)$,

$$\lim_{n\to\infty} \frac{1}{n} Q^{\varepsilon}_{\mathrm{retro}}(\mathcal{N}^{\otimes n}) = \frac{1}{2} \lim_{n\to\infty} \frac{1}{n} C^{\varepsilon}_{\mathrm{retro}}(\mathcal{N}^{\otimes n}) = \frac{1}{2}\left(I_{\max}(\mathcal{N}) + I^{\infty}_{\mathrm{doe}}(\mathcal{N})\right)$$

regularization (non-additive)

Max-information: $I_{\max}(\mathcal{N}) := \inf_{\lambda, \sigma_B} \left\{ \log_2 \lambda : \mathcal{N}_{A\to B}[\Phi_{A'A}] \leq \lambda \frac{I_{A'}}{d_{A'}} \otimes \sigma_B \right\}$ König et al., IEEE TIT 55, 4337 (2009)

Doeblin information: $I_{\mathrm{doe}}(\mathcal{N}) := \inf_{\lambda, \tau_B} \left\{ \log_2 \lambda : \lambda \frac{I_{A'}}{d_{A'}} \otimes \tau_B \leq \mathcal{N}_{A\to B}[\Phi_{A'A}] \right\}$ George et al., arXiv:2503.22823 (2025)

unit-trace Hermitian operator!

Optimal strategy: Amplified probabilistic teleportation

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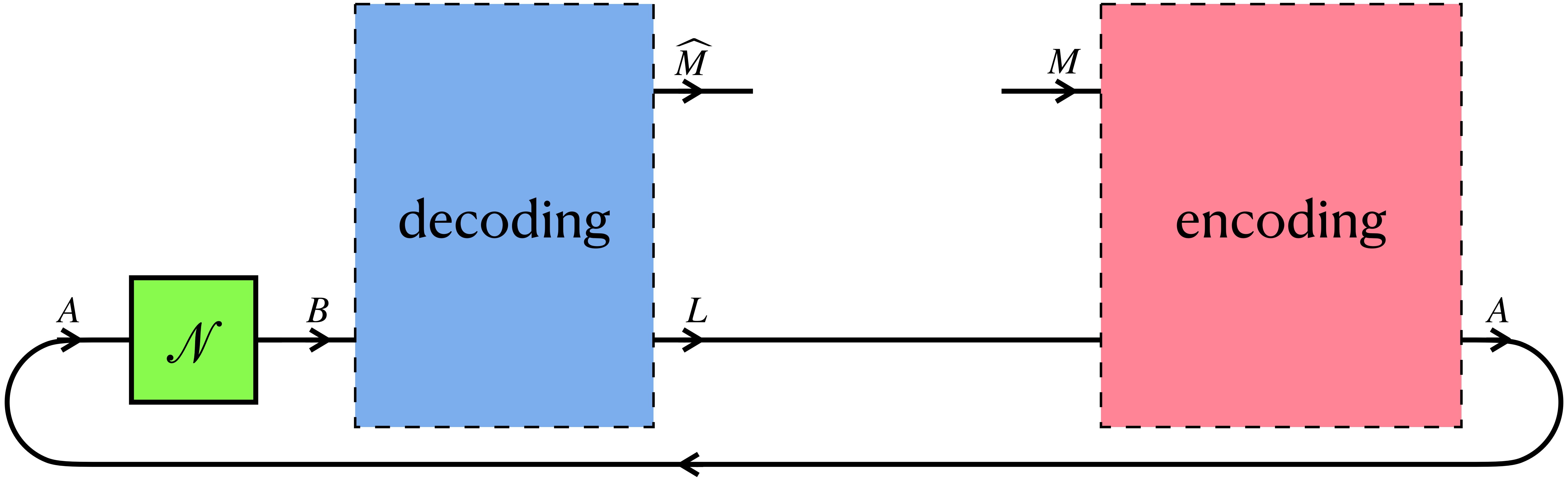
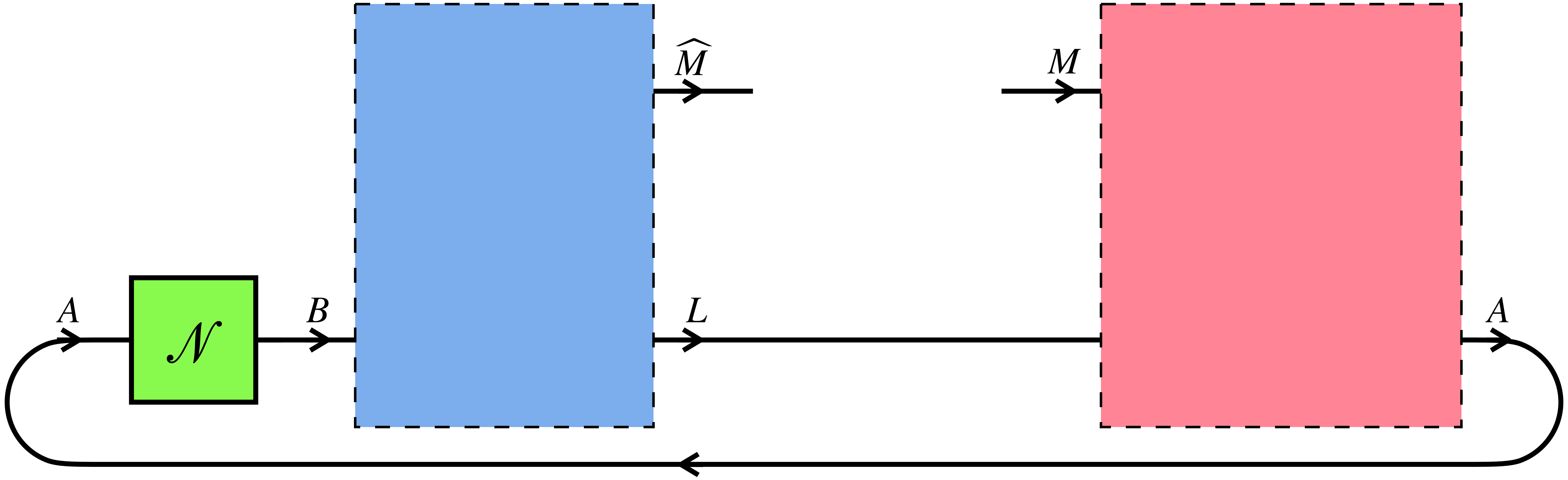


Optimal strategy: Amplified probabilistic teleportation

Instead of encoding the message in the input of the noisy P-CTC, the sender tries to send the message via probabilistic teleportation through $M \to L_1 \to \widehat{M}$



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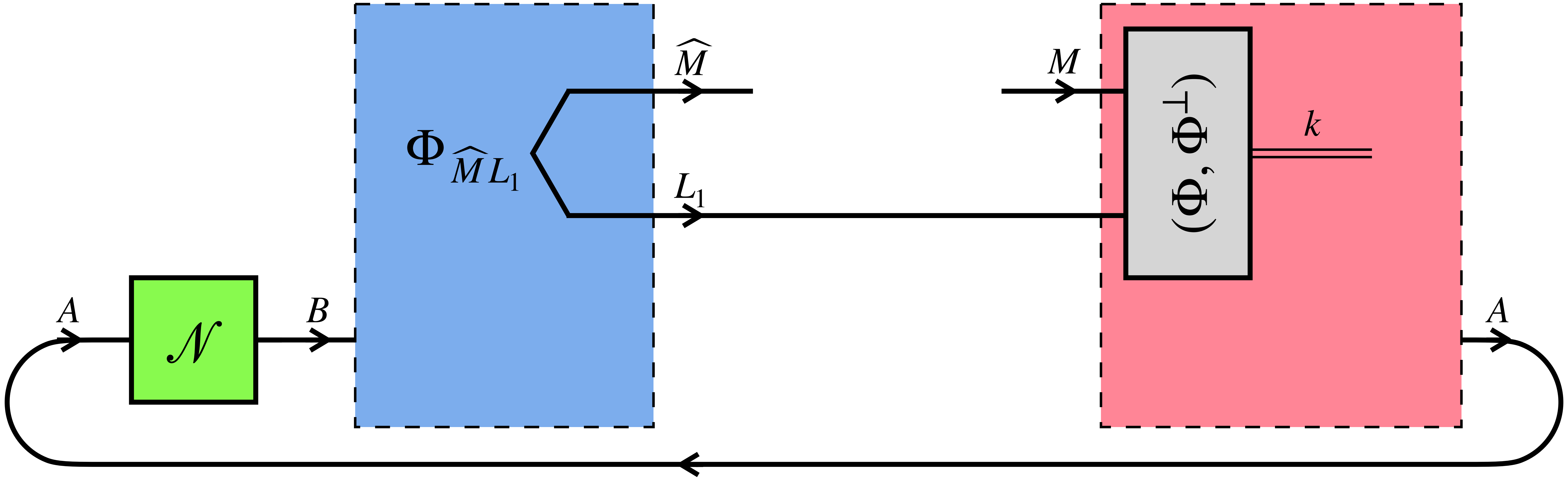

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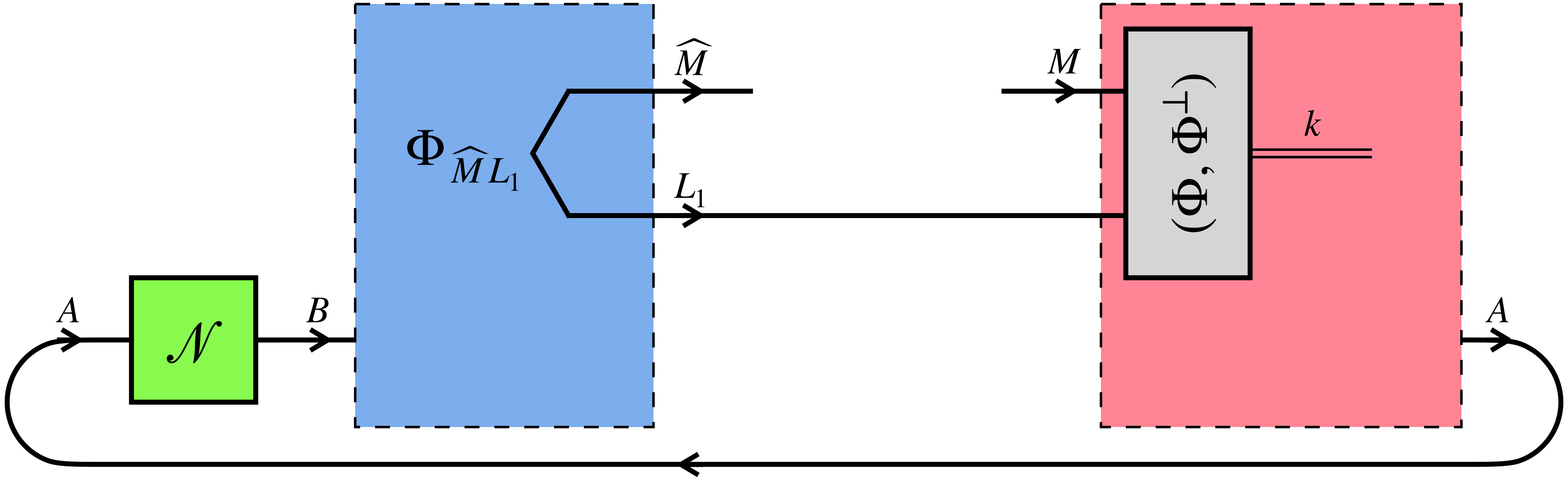
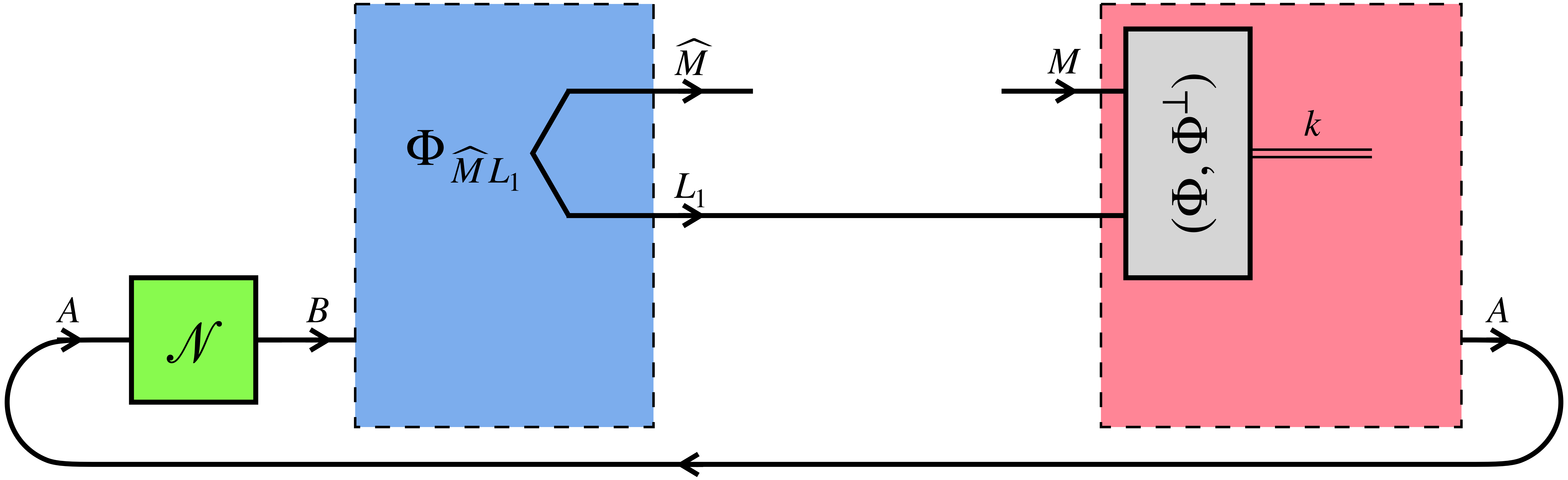


- The probability of success ($k = 0$) is of course only $1/d_M^2$

Optimal strategy: Amplified probabilistic teleportation

Engineer a causal loop consisting of $\mathscr{N}$ and a channel $\mathscr{K}^{(k)}$, conditioned on the outcome $k \in \{0,1\}$ of the probabilistic teleportation



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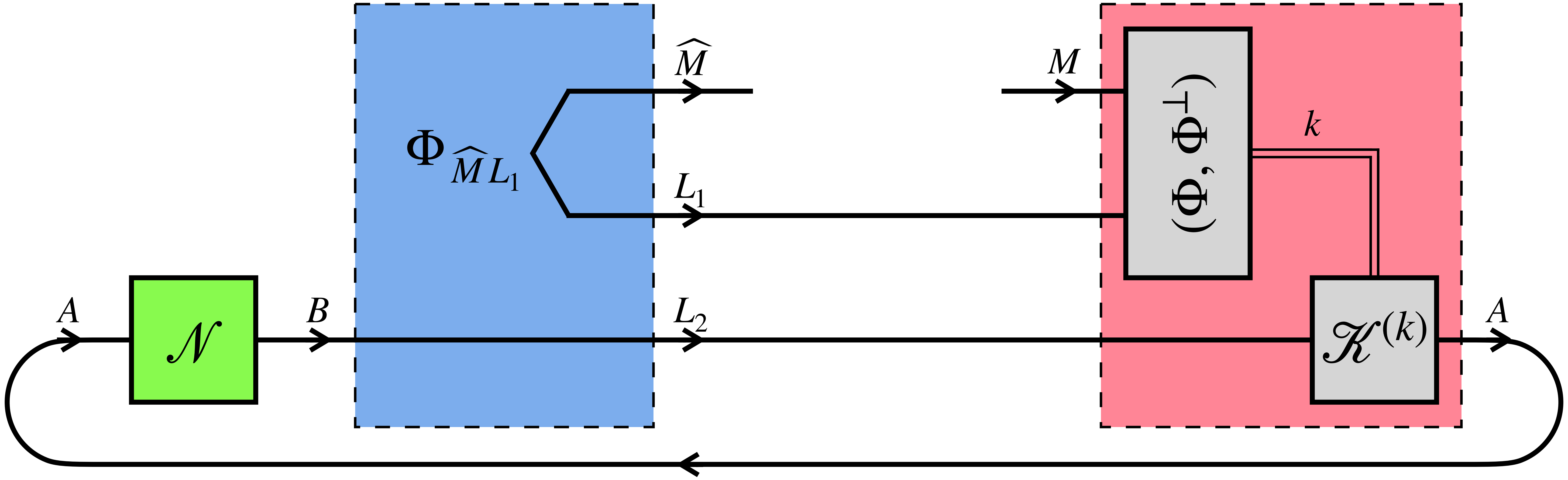

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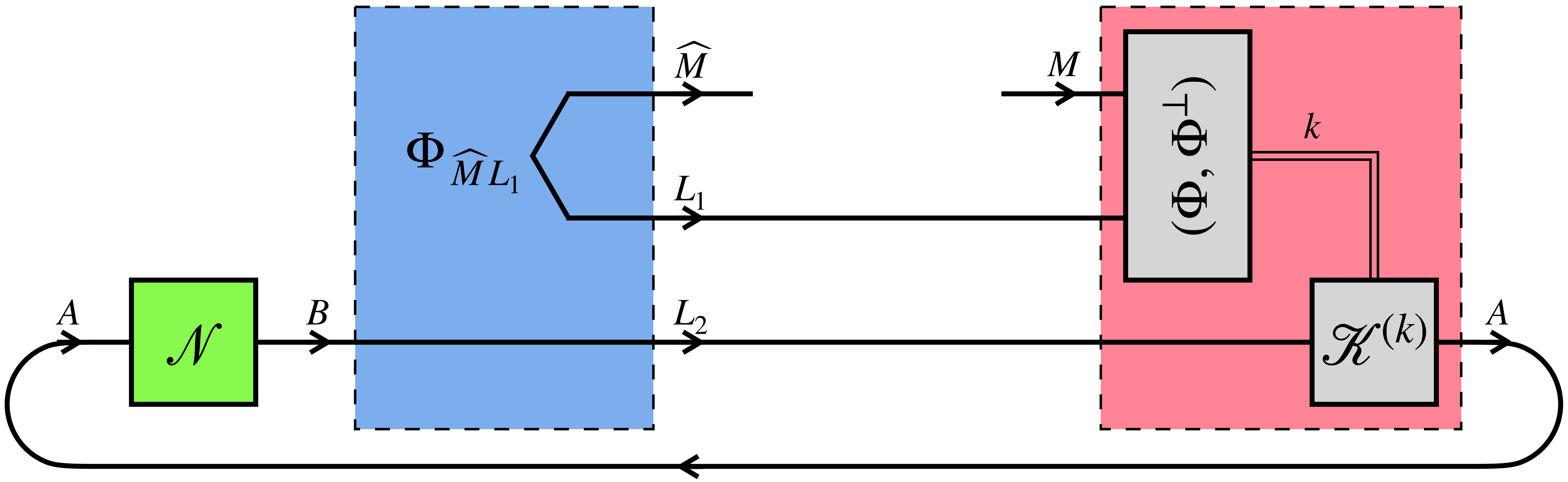

- Due to nonlinearity, the odds of a successful teleportation are amplified from $\dfrac{1/d_M^2}{1-1/d_M^2}$ to $\dfrac{1/d_M^2}{1-1/d_M^2} \cdot \dfrac{\mathrm{Tr}[\Phi_{AA'}(\mathscr{K}^{(0)}_{B\to A} \circ \mathscr{N}_{A\to B})[\Phi_{AA'}]]}{\mathrm{Tr}[\Phi_{AA'}(\mathscr{K}^{(1)}_{B\to A} \circ \mathscr{N}_{A\to B})[\Phi_{AA'}]]}$

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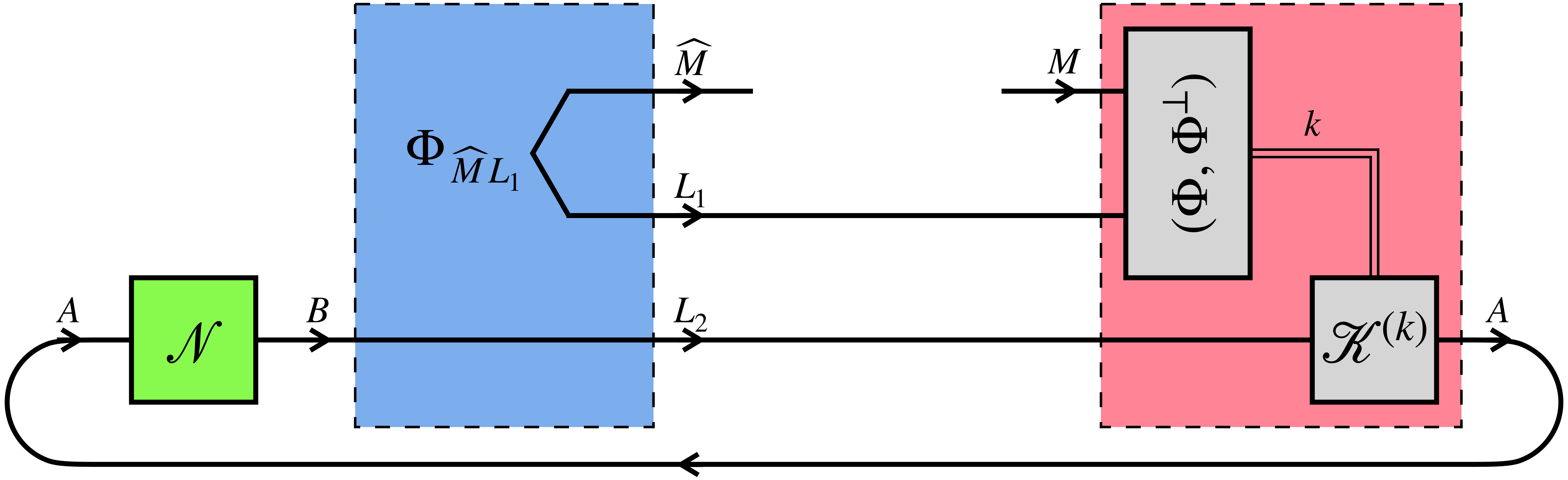

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Choose $\mathscr{K}^{(0)}$ and $\mathscr{K}^{(1)}$ such that the above odds are maximized

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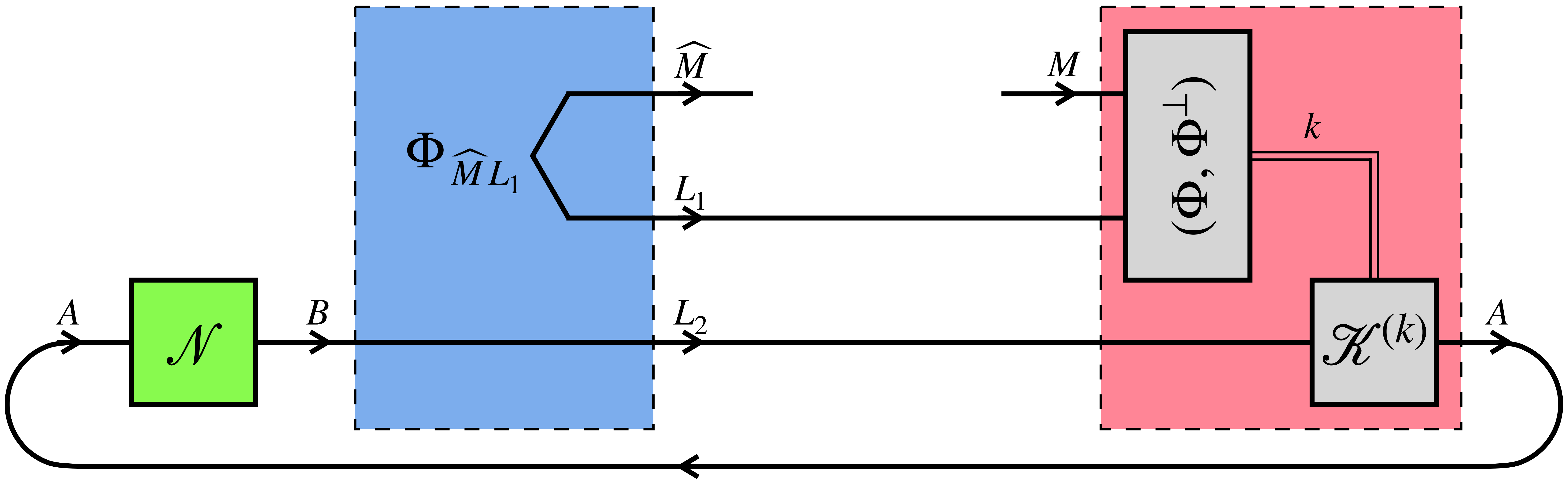

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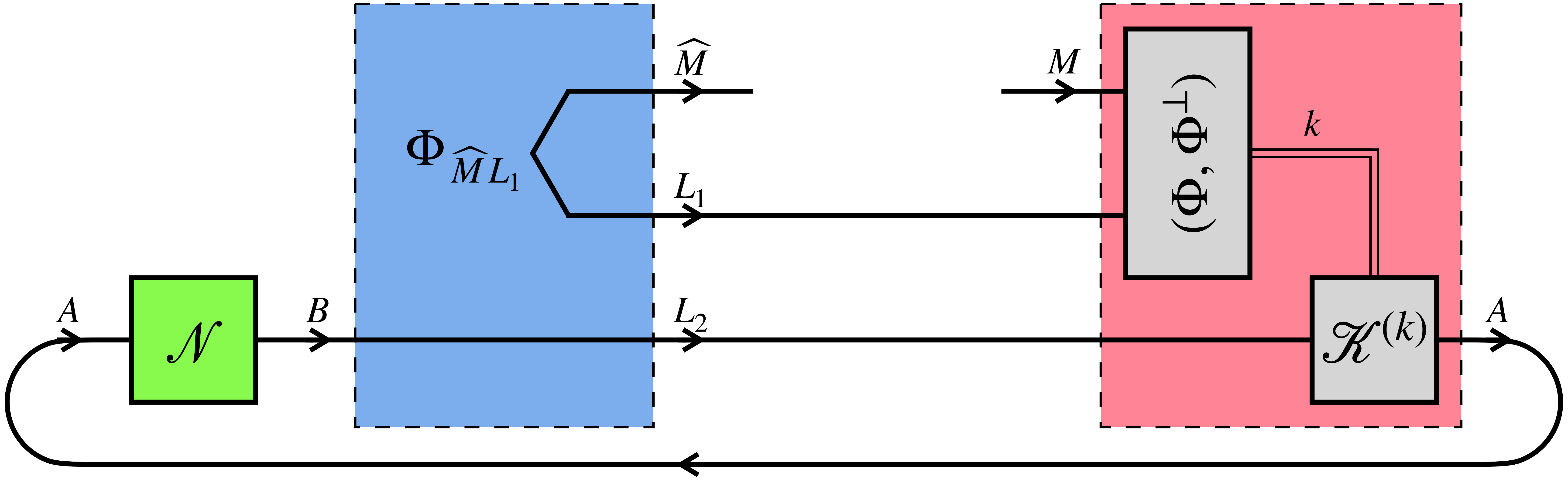

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- When $\log_2 d_M =$ the one-shot retrocausal quantum capacity, the maximized odds $\geq \frac{1-\varepsilon}{\varepsilon}$

Conclusion

Retrocausal communication: messaging the past from the future through a given noisy P-CTC

- Solved one-shot retrocausal quantum & classical capacity (closed-form, SDP-computable)
- Noisy-“time-machine” coding theorem (backward version of noisy-channel coding theorem)

$$\langle \overleftarrow{\mathcal{N}} \rangle + \infty[q \to q] \geq \frac{1}{2}\left(I_{\max}(\mathcal{N}) + I^{\infty}_{\mathrm{doe}}(\mathcal{N})\right)[q \leftarrow q],$$

noiseless time machine

noisy time machine

noiseless channel

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Open questions:

- What if the memory that the daughter leaves for the father is noisy?

$$\langle \overleftarrow{\mathscr{N}} \rangle + (?)\langle \overrightarrow{\mathscr{L}} \rangle \geq (?)[q \leftarrow q],$$

- Connection with black-hole final-state models?

Horowitz and Maldacena, JHEP 2004, 008 (2004)

Conclusion

arXiv:2509.08965

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